PEARL Power Electronics Laboratory – Final Report

Characterization, Analysis and Verification of a PFC Flyback Converter

Final Report

Submitted by: Anat Rom, Shoval Buzaglo

Supervisor: Eric Herbelin

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1. Introduction

This project deals with the design, analysis and verification of a power-factor-corrected Flyback converter of our own design, in which an Arduino-class microcontroller replaces the dedicated analogue PFC controller. The starting point of the work was an existing commercial unit, the PFC Flyback controller EV44010-S+4688-N-01A, which we characterized and analysed in order to establish the reference behaviour that a correct design must reproduce. On that basis we designed our own 12 V / 9 W converter and built two complete models of it – one without power factor correction and one with it – which are the subject of all the results reported here.

Throughout the project the influence of the operating and loading conditions on the power factor, on the input-current distortion and on the stability of the output voltage is examined. The central research hypothesis is that a correctly designed PFC stage fully decouples the character of the load from the utility grid, so that any change in power factor should originate from a change in the power level alone and not from the character of the load itself (resistive, inductive, capacitive or pulsed). The second question the project asks is whether such a stage can be realised with a general-purpose microcontroller instead of a dedicated PFC chip, and at what cost in performance.

The project was carried out in two stages. In the first stage the existing commercial PFC controller was studied and characterized and used as the design reference; that study is documented in Chapters 5 to 7. In the second stage we designed our own converter, described in Chapter 8, and simulated both of its variants in full. Every simulation result presented in this report was obtained from our own models, and the laboratory measurement campaign described in Chapter 4 is the next phase of the work.

2. Theoretical Background

Many modern electronic systems present a load that is neither linear nor continuous. Switching circuits (a camera flash, for example) draw current in violent peaks rather than continuously, and DC motors generate significant voltage and current noise on the DC rail. In a system that is not properly designed, loads of this type cause serious distortion of the mains supply.

The purpose of the PFC controller is to prevent this, by ensuring that the input current follows the shape of the input voltage. In a correctly operating PFC system the input current is kept almost perfectly sinusoidal and in phase with the voltage. In order to achieve this despite the disturbances coming from the load, the PFC control loop is designed to be very slow (a bandwidth of roughly 10 Hz). This means that the controller does not attempt – and is not able – to respond to the fast pulses of the output load. Instead, it averages the required power and draws it from the grid in a smooth, sinusoidal manner.

In practice, the instantaneous energy required by pulses and fast load changes is supplied by the large DC capacitors of the circuit, which absorb the "blows" of the load. This design, together with the isolating transformer and the control strategy, forms a "wall" that effectively isolates the "dirty" DC side from the "clean" AC mains.

2.1 Analytical model and performance metrics

The evaluation of system performance rests on a basic analytical model. The central figure of merit is the Total Harmonic Distortion (THD) of the input current; the true power factor is the product of the displacement factor and the distortion factor; the input power is obtained from the cycle-averaged product of instantaneous voltage and current; and the efficiency of the system is measured as the ratio between the useful output power and the input power. The four quantities used throughout this report are defined as follows:

- True power factor – $PF = \frac{P_{in}}{S_{in}}$, where P_{in} is the cycle-averaged real input power and $S_{in} = V_{rms} \cdot I_{rms}$.
- Displacement factor – $cos(\varphi_1)$, the cosine of the phase difference between the fundamental components of the input voltage and the input current.
- Distortion factor – $\frac{I_{1,rms}}{I_{rms}}$, expressing the fraction of the RMS current that is carried by the fundamental; PF = displacement factor × distortion factor.
- Total harmonic distortion – $THD = \frac{\sqrt{\Sigma_{\{n\geq 2\}} I_n^2}}{I_1}$, evaluated here over harmonics 2 to 40, as required by the harmonic-emission standards.
- Efficiency – $\eta = \frac{P_{out}}{P_{in}}$, with P_{out} taken as the total useful power delivered to all output loads.

In order to test these principles, the research approach requires the DC output to be "abused": a pure resistive load is replaced by reactive loads, sharp current steps and repeated pulses, in order to demonstrate that despite the extreme variations the input side remains indifferent and continues to present a high power factor (above 0.95). This is done first in simulation, on the two models of Chapter 8, and afterwards on the hardware prototype.

3. Key Performance Targets

In order to verify the performance of our converter, and its ability to isolate the mains from the character of the load as well as the reference unit does, the following key performance targets were defined – first as acceptance criteria for the simulation, and then for the laboratory measurements:

- High and stable power factor: the central target is to demonstrate that the input stage presents a power factor close to unity. As a baseline reference under a purely ohmic load, the PF is expected to be 0.98 or better. Under varying loads (dynamic, pulsed and reactive) the target is to show that the PF remains indifferent and stays above 0.95.
- Current–voltage tracking and a verified switching regime: proof that the input current remains almost sinusoidal and in phase with the input voltage, and identification of the conduction regime the converter actually works in – for our fixed-frequency, constant-on-time design this means verifying that the transformer demagnetises fully in every switching cycle and that the commanded on-time stays constant along the line cycle.
- Low total harmonic distortion: this is a central metric in PFC systems. The target is to measure and compare the THD values under different load levels (light, nominal and heavy) and different load types, and to show that the input stage exhibits at most a mild increase in THD under "violent" loads, while keeping the AC mains clean.
- Output-voltage stability and transient response: examination of the behaviour of the output voltage under load steps. The goal is to measure the instantaneous voltage excursion and the recovery of the system, and to check to what extent the event affects the PF.
- Energy efficiency: measurement of converter efficiency under the various operating conditions, and identification of where the losses are incurred.
- Feasibility on a microcontroller: demonstration that the whole control law fits the arithmetic and timing budget of an Arduino-class device – a slow software loop and one timer compare register, with no fast current regulator and no zero-current interrupt.

4. Design Methodology and Work Plan

The project is based on an ordered methodology that starts with theoretical research and the characterization of an existing reference unit, proceeds through the design of our own converter and its simulation, and ends with a laboratory measurement campaign on the hardware prototype. The work plan comprises the following stages:

Stage 1 – Theoretical study and requirement definition

A review of the professional literature on the Flyback converter, on current-mode and constant-on-time control and on the principles of power factor correction. At this stage the central research question is defined and the research hypothesis is set: proving the independence of the power factor from the character of the output load.

Stage 2 – Characterization of the reference unit

Study of the existing PFC Flyback controller EV44010-S+4688-N-01A: collection of its technical data (input and output voltages, nominal power, switching behaviour), analysis of its internal control loop and of the way it enforces boundary conduction mode, and identification of the measurement points in the circuit. This stage produces the reference behaviour and the analytical expressions for input power, efficiency and THD against which our own design is judged. It is documented in Chapters 5 to 7.

Stage 3 – Design of our converter

Specification and design of a 12 V / 9 W isolated Flyback converter controlled by an Arduino-class microcontroller: choice of topology and magnetics, sizing of the switch, the sense resistor, the clamp and the output stage, design of the isolated TL431/optocoupler feedback path, and – the central decision – choice of the control law that can shape the input current within the arithmetic budget of the processor. Two variants were designed on a common power stage, one without PFC and one with it. This stage is documented in Chapter 8.

Stage 4 – Modelling and simulation campaign

Construction of complete transistor-level LTspice models of both variants, including behavioural models of the firmware arithmetic, and execution of a systematic set of simulations: verification of the switching regime, a no-PFC baseline, the three static load characters, and four magnitudes of repeated load step. The waveforms are post-processed numerically to extract power factor, displacement and distortion factors, harmonic spectra, efficiency and output-voltage excursion. This stage is documented in Chapters 9 and 10.

Stage 5 – Design of the measurement system

Construction of the laboratory environment using a regulated AC source connected to a power analyzer at the input. On the output (DC) side an advanced electronic load capable of dynamic operation is connected, together with an oscilloscope and current probes for continuous measurement of the variables, and the measurement points are matched to the nodes used in the simulation so that the two data sets are directly comparable.

Stage 6 – Laboratory measurements on the prototype

The hardware prototype is examined under the same series of controlled experiments used in simulation: a classic resistive load as baseline; reactive loads at the same power level; sharp repeated current steps of increasing magnitude to emulate pulsed and switched behaviour; and a separation of variables, in which power changes are performed decoupled from changes of load type, in order to prove that if an effect on the PF exists it originates from the total power and not from the character of the load.

Stage 7 – Data processing and conclusions

Collection of the data and creation of complete measurement tables for PF, THD and efficiency under each load condition, presentation of the results through comparative graphs, and comparison of three data sets against one another: the theoretical prediction, the simulation of our models, and the laboratory measurements – leading to the final conclusion and to the confirmation or refutation of the research hypothesis.

5. Conduction Modes and the BCM Regime

The operating regimes of switched-mode converters are determined by the behaviour of the inductor current (or, in the case of a Flyback converter, of the transformer current) during the switching cycle. In order to understand the BCM (Boundary Conduction Mode) regime in depth, it is first necessary to define the two modes between which it lies: CCM and DCM.

5.1 Continuous Conduction Mode (CCM)

In this regime the inductor current never falls to zero during the switching cycle. The current oscillates around an average value (the load current), with a ripple determined by the input voltage, the output voltage, the switching frequency and the inductance.

- Advantages: relatively low peak currents, which reduces the conduction losses in the components and allows the use of smaller output filter capacitors.
- Disadvantages: high switching losses, since the switch turns on while a high current still flows through the diode (reverse-recovery phenomenon). In addition, in certain converters (such as the Flyback) the control of the system is more complex because of the presence of a right-half-plane zero (RHPZ) in the transfer function.

5.2 Discontinuous Conduction Mode (DCM)

In this regime the inductor energy is completely delivered to the load before the end of the switching cycle. As a result, the inductor current falls to zero and remains at zero for a certain interval (dead time) until the beginning of the next cycle.

- Characteristics: appears typically at light loads or when the inductance is small.
- Advantages: no reverse-recovery problem in the diode (it turns off at zero current). In Flyback converters the transfer function becomes a first-order system at low frequencies, which greatly simplifies the design of the control loop (no RHPZ).
- Disadvantages: peak and RMS currents that are very high relative to the average current, which leads to higher conduction losses and requires components with a higher current rating and a more massive EMI filter.

5.3 Boundary Conduction Mode (BCM)

The BCM regime (sometimes also called CrCM – Critical Conduction Mode) is exactly the seam line between CCM and DCM. In this regime the inductor current falls exactly to zero, and at that very instant the main switch is turned on again and starts the next cycle. There is no interval in which the current remains at zero (as in DCM), and the cycle does not start from a positive current (as in CCM).

Unlike CCM and DCM, which usually operate at a fixed switching frequency, BCM systems normally operate at a variable switching frequency (variable-frequency control). A detailed description is given in Chapter 7 – the MP44010 controller.

- Advantage – soft turn-on (ZCS/ZVS): since the switch is turned on exactly when the current is zero, the turn-on losses are almost eliminated.
- Advantage – no reverse recovery: the diode always turns off at zero current, which completely eliminates reverse-recovery losses and allows the use of cheaper diodes (expensive SiC diodes, for instance, are not mandatory).
- Advantage – moderate peak currents (relative to DCM): the peak current in BCM is exactly twice the average load current. This is indeed higher than in CCM, but significantly lower than the violent currents of DCM.
- Disadvantage – variable frequency: because the switching frequency varies with the load and with the instantaneous voltage, the design of the EMI filter becomes more challenging, since the switching noise is spread over a wide frequency spectrum.
- Disadvantage – power limitation: because the peak current is twice the average current, BCM is suited mainly to low and medium power levels (typically 300 W–400 W). At higher powers the peak current creates excessive conduction losses, and it is customary to move to CCM.

6. The Reference Unit: Flyback Power Stage

This chapter and the next document the existing unit that served as our design reference. The Flyback converter used in it as the power-factor-correction stage is based on several key components that enable energy storage, transformation and regulation of the output voltage. Unlike a classic Flyback converter fed from a fixed DC voltage, this converter is fed from a fully rectified AC line voltage by means of a diode bridge, and therefore its input voltage varies as a rectified sine wave over time.

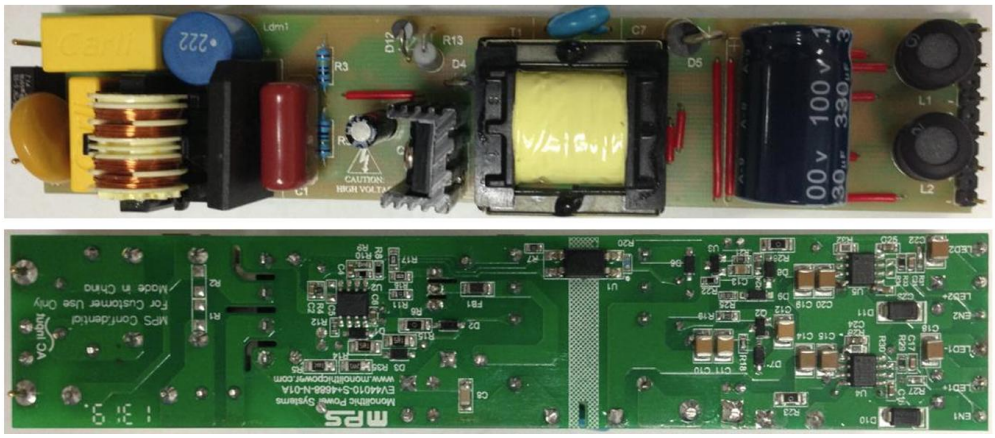


Figure 1a – Structure of the given Flyback converter.

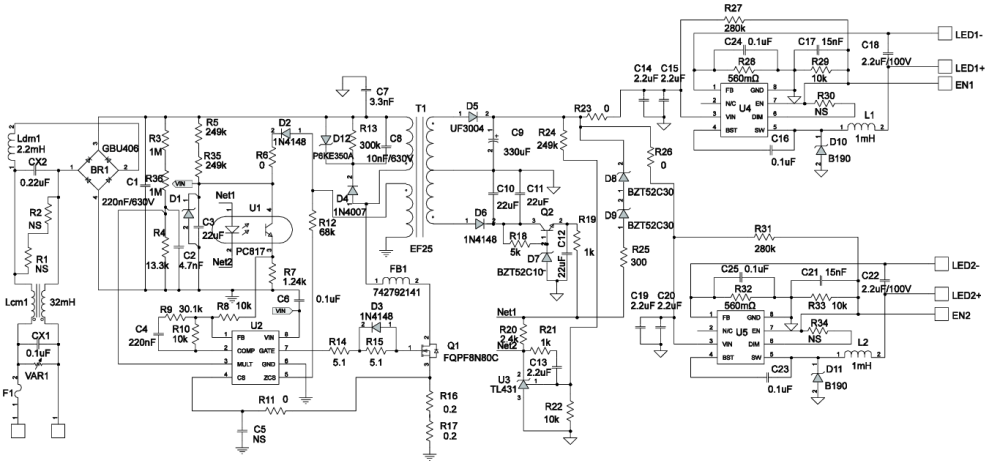


Figure 1b – Schematic of the given Flyback converter.

7. The Reference Unit: the MP44010 Controller

The MP44010 controller, which is the control device inside the reference unit, is a dedicated power-factor-correction device operating in Boundary Conduction Mode. The controller combines several internal blocks that allow optimal closed-loop control of the output voltage while keeping the input current sinusoidal and in harmony with the mains voltage. Understanding it in detail is what allowed us to decide, in Chapter 8, which parts of this control strategy to reproduce digitally and which to replace.

7.1 Controller structure and pin functions

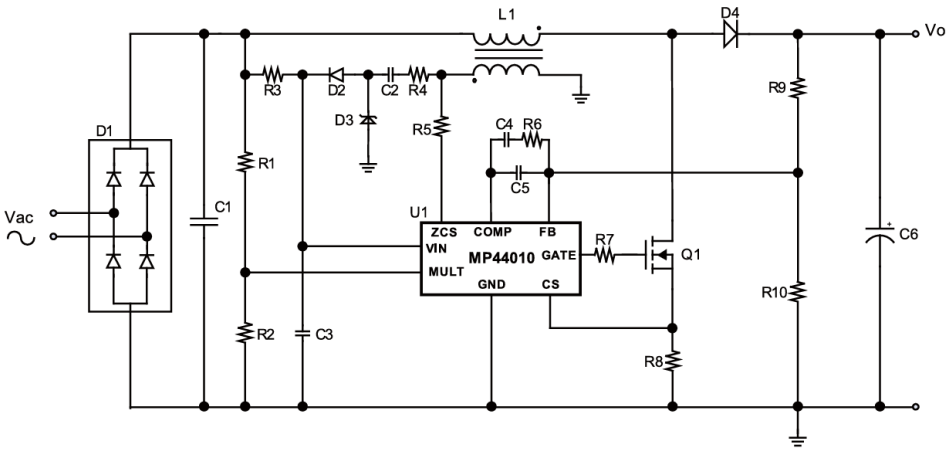


Figure 2 – Typical structure of the control loop to which the MP44010 controller is connected.

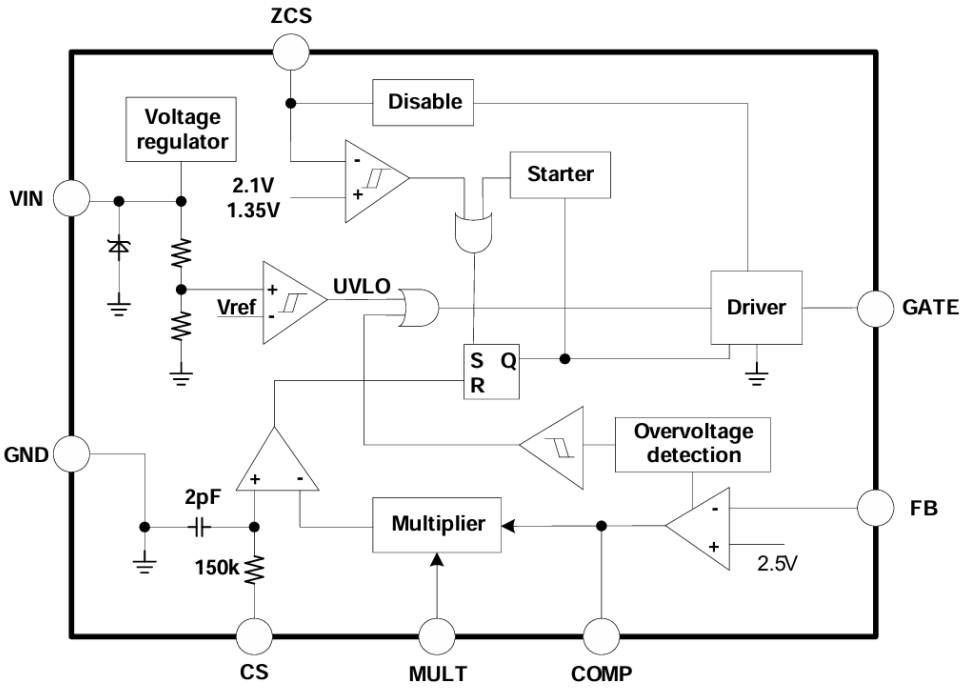


Figure 3 – Block diagram of the MP44010 controller, the PFC controller inside the Flyback converter.

The structure of the controller is based on eight central pins, each of which represents an essential function in the control scheme:

- FB (Pin 1) and COMP (Pin 2) – the inputs of the internal error amplifier. The FB pin samples the regulated output voltage through a resistive divider and compares it to a precise internal reference. A compensation network connected between FB and COMP dictates the dynamics of the loop. Because of the requirements of PFC, the bandwidth of the voltage loop is designed to be significantly lower than the line frequency (below ~10 Hz), in order to prevent correction of the double-line-frequency ripple, and thus to keep the COMP voltage constant over a half line cycle.
- MULT (Pin 3) – the multiplier input. This pin is connected to the rectified AC line voltage through a resistive divider and provides the controller with the sinusoidal mirror image of the input voltage.
- CS (Pin 4) – the current-sense pin. The voltage developed across a sense resistor connected in series with the source of the main MOSFET is fed to this pin. The device contains an internal R/C filter built into this pin, for filtering high-frequency switching noise.
- ZCS (Pin 5) – the zero-current-detector input. The pin is connected to an auxiliary winding of the transformer and detects the precise instant at which the current in the winding reaches zero, which serves as the trigger for the start of a new switching cycle.
- GATE (Pin 7) – the gate-driver output, capable of supplying peak sourcing and sinking currents for fast and efficient switching of the MOSFET.
- VIN (Pin 8) and GND (Pin 6) – the supply and ground pins of the device. A minimum supply voltage is required for correct operation.

7.2 The control mechanism and the enforcement of BCM in a Flyback converter

In a Flyback converter designed for PFC, the goal is to make the average value of the input current follow accurately the waveform of the input voltage. The MP44010 achieves this through a combination of constant

on-time control and the enforcement of operation exactly at the transition point between continuous conduction (CCM) and discontinuous conduction (DCM) – that is, the BCM regime.

(a) The on-time phase and the creation of a sinusoidal envelope

At the instant the MOSFET is turned on, the instantaneous rectified input voltage appears across the primary winding of the Flyback transformer. The input current (which is the primary winding current) begins to rise linearly from zero at a rate set by the ratio of the instantaneous input voltage to the primary inductance. The current flows through the sense resistor and produces at the CS pin a voltage that rises linearly together with the winding current. In parallel, the internal multiplier receives two signals: the rectified sine reference from the MULT pin and the slow error voltage from the COMP pin. The multiplier output produces a sinusoidal reference voltage that varies along the line cycle.

The instant at which the CS voltage reaches exactly this reference value constitutes the turn-off condition. At that moment the internal current comparator toggles, resets the SR latch, and the gate driver turns the MOSFET off. Since the COMP voltage is almost perfectly constant over a half line cycle, the peak current of the primary winding is bounded by a perfect sinusoidal envelope. Because the on-time is determined by the ratio between the reference and the input voltage slope – a ratio kept constant by the multiplier – a constant on-time is obtained for a given operating point of output voltage and current.

(b) The off-time phase and zero-current detection (ZCS)

At the turn-off instant of the MOSFET, the energy stored in the magnetic field of the transformer is transferred to the secondary winding, the output diode becomes forward biased, and the secondary current begins to fall linearly towards the output.

During this phase the primary and secondary winding voltages are also reflected onto the auxiliary winding connected to the ZCS pin. As long as current flows in the secondary, the voltage on the auxiliary winding remains positive and above the upper threshold of the ZCS block. At the precise instant at which the secondary current reaches zero (complete depletion of the energy stored in the transformer), the voltage across the transformer windings collapses and reverses, because of the onset of resonant oscillations with the MOSFET capacitance. The ZCS detector identifies this voltage drop. As soon as the voltage at pin 5 falls below the lower threshold (falling edge), the controller immediately recognises this as the zero-current point, generates an internal set pulse to the SR latch and turns the MOSFET on again for the next cycle.

(c) How the BCM regime guarantees a high power factor

As a result of this operation, the next switching cycle begins exactly at the point at which the current has reached zero. The current waveform in the winding is triangular in every switching cycle, its minimum value is always zero, and its peak follows the mains sine wave. The average current drawn from the mains in each switching cycle equals one half of the instantaneous peak current.

Since the peak current is sinusoidal and perfectly in phase with the mains voltage, the average current is also sinusoidal and perfectly in phase with the input voltage. This fact guarantees a power factor very close to unity and minimal harmonic distortion, as required by strict international standards such as IEC 1000-3-2.

7.3 Built-in protection mechanisms and performance enhancement

Beyond the main control loop, the MP44010 integrates advanced functional blocks that improve the reliability of the system:

- Two-stage over-voltage protection (OVP): PFC converters suffer from an inherent slowness of the voltage loop. In the event of a sudden load dump, the output voltage may rise rapidly. The device protects the system in two ways. Dynamic OVP: the controller monitors the current injected into the COMP pin through the compensation capacitors as a result of a fast change of the FB voltage; if this current exceeds a first threshold the multiplier output is forced down, and if it reaches a second threshold the gate driver is blocked completely and the device enters an idle state until the voltage decays. Static OVP: at very light load, or with no load at all, the COMP voltage falls below a low saturation threshold of the error amplifier; the controller detects this, blocks the gate and moves to burst-mode operation, in order to prevent a rise of the output voltage and to save energy.
- Zero-crossing compensation: close to the zero crossing of the mains voltage, the existing on-time is not sufficient to drive a significant current into the winding because the input voltage is very low, which causes zero-crossing distortion and raises the THD. The controller offers an internal mechanism that injects a built-in voltage offset into the multiplier output near the zero crossing. This slightly extends the on-time in these regions, forces an increased energy transfer at the bottom of the sine wave, and lowers the THD appreciably. The offset gradually vanishes as the instantaneous input voltage rises towards the peak, in order to prevent an excess of energy.
- Internal starter: when the system is powered up there is as yet no current and no switching in the converter, and therefore the auxiliary winding does not generate a signal for the ZCS pin. In order to solve this chicken-and-egg problem, the controller includes an internal start-up timer. If no signal is received at the ZCS pin within this time, the timer generates a deliberate turn-on pulse to the gate, which initiates the initial switching process and the flow of current in the winding.

The combination of the features of the MP44010 makes it possible to design an efficient PFC Flyback converter with a small number of external components (thanks to the filters and timers integrated in the chip), which offers high efficiency at full load and full compliance with modern power-quality requirements. Our own design, described in the next chapter, keeps the essential idea of this controller – a slow voltage loop commanding a constant on-time – while dispensing with the multiplier, the zero-current detector and the auxiliary winding, none of which a microcontroller can support at switching speed.

8. Our Design – an Arduino-Controlled Flyback Converter

The study of the existing EV44010-S+4688-N-01A unit described in Chapters 6 and 7 served as the reference for the main part of the project: the design of our own 12 V / 9 W Flyback converter, in which the dedicated analogue PFC controller is replaced by an Arduino-class microcontroller acting as the control unit. Two complete models were built and simulated in LTspice, sharing an identical power stage and differing only in the control law: a conventional model without power factor correction, and a power-factor-corrected model. All the simulation results presented in Chapter 10 are results of these two models of our own design, not of the commercial unit.

8.1 Design specification

- Input: single-phase AC, 35.3553 V RMS (50 V peak) at the line frequency, applied through a differential EMI filter.
- Output: 12 V DC, nominally 9 W (approximately 0.7 A), galvanically isolated from the mains.
- Topology: single-switch Flyback converter fed from a full-wave diode bridge.
- Controller: a microcontroller (Arduino) generating the gate command; modelled in LTspice as a behavioural control block that reproduces exactly the arithmetic the firmware performs.

- Feedback: isolated output-voltage sensing through a TL431 programmable shunt reference driving a 4N25 optocoupler.
- Two variants: a conventional fixed-frequency variant without PFC, and a fixed-frequency constant-on-time variant with digital PFC – both operating in discontinuous conduction mode, for the reasons set out in Section 8.7.

8.2 The common power stage

Both models share the same power path, so that any difference in the results can be attributed to the control law alone. Starting at the mains terminals (nodes AC_L and AC_N), the current passes through a differential-mode EMI filter consisting of a series inductor Lf with a modelled winding resistance of 0.3 Ω and a shunt X-capacitor Cx with a modelled ESR of 0.5 Ω. The filtered AC is rectified by a full-wave diode bridge, and the rectified voltage feeds the primary winding of the Flyback transformer.

The primary winding is switched by a MOSFET whose source returns to ground through a current-sense resistor (RSENSE). The voltage across this resistor is the converter's current measurement; it is passed through an R–C filter to the node CSF, which removes the leading-edge spike and the switching noise so that the controller – whether analogue comparator or microcontroller ADC – sees a clean image of the primary current. On the secondary side, a Schottky diode rectifies the transformer current into the output capacitor bank, which forms the 12 V rail between nodes out and COM and supplies the load R_LOAD.

Regulation is closed across the isolation barrier by a TL431 shunt reference on the secondary side, which compares the output voltage through a resistive divider against its internal 2.5 V reference and sinks current through the LED of a 4N25 optocoupler. The phototransistor on the primary side converts that current back into the control voltage seen by the controller. This arrangement keeps the safety isolation intact while giving the primary-side controller an accurate measure of the output error.

8.3 The conventional model – no power factor correction

In the model without PFC the microcontroller runs a straightforward voltage-mode regulation loop: it drives the MOSFET at a fixed switching frequency and adjusts only the duty cycle, according to the error signal arriving from the optocoupler. The control bandwidth is high enough to hold the 12 V rail steady, and no attempt whatsoever is made to shape the input current.

The consequence is the behaviour familiar from any uncorrected offline supply. A bulk capacitor after the bridge holds the rectified peak, so the bridge diodes conduct only during the short interval in each half-cycle in which the instantaneous line voltage exceeds the capacitor voltage. The converter therefore draws its energy in narrow, high-amplitude current pulses located at the crest of the voltage. This model is the baseline of the study, and its measured behaviour is reported in Sections 10.1 and 10.2.

8.4 The PFC model – digital constant-on-time control

The PFC model keeps the same power stage but replaces the duty-cycle regulator with a constant-on-time control law: the switching frequency stays fixed at 100 kHz and the controller regulates the length of the on-pulse instead of its duty ratio. Four deliberate changes turn the conventional converter into a power-factor-corrected one:

- The bulk capacitor is removed. Cin falls from 125 µF to 10 nF, so the rectified rail is no longer held at the peak but follows the rectified sine instantaneously. This single change is what makes power factor correction possible at all: the bridge now conducts throughout the whole half-cycle instead of only at the crest.

- The feedback path is deliberately slowed. The optocoupler signal is passed through Rfb = 100 kΩ and Cfb = 150 nF to the node Vfb_slow, a first-order low-pass with a corner at $\frac{1}{2\pi RC} \approx 10.6\,Hz$ – an order of magnitude below the line frequency, so the loop reacts to the average output power and is blind to the double-line-frequency ripple.
- The controlled variable becomes the on-time. The firmware computes TON_CMD = limit(Ton0 + Kp·e + Ki·∫e dt, TonMin, TonMax) with Ton0 = 3.0 µs, Kp = 0.1, Ki = 20 and clamps at 0.15 µs and 4.2 µs, where e is the error between the 2.5 V reference and the scaled output measurement Vadc = Vout·2.5/12.
- The comparison ramp is scaled to time. The sawtooth rises from 0 V to 10 V over the 10 µs switching period – exactly 1 V per microsecond – so comparing TON_CMD against it converts the command voltage directly into an on-time in microseconds, which is precisely what a microcontroller timer compare register does in hardware.
- Protection is retained. The latch is also reset if the filtered current sense V(CSF) exceeds VcsLim = Rsense · Ilim = 0.14 Ω × 2.6 A = 0.364 V, giving cycle-by-cycle peak-current limiting on top of the on-time law.

This is the central design decision of the project. A conventional average-current-mode PFC would require the microcontroller to close a fast current loop – sampling the line voltage, multiplying it by the error signal and regulating the current against that reference at switching speed – which is far beyond the arithmetic budget of an Arduino-class device. Constant-on-time control moves the multiplication into the physics of the transformer. Because the primary current rises from zero at a rate Vin(t)/Lp, a fixed on-time makes the peak current of every switching cycle proportional to the instantaneous line voltage, and the cycle-averaged input current becomes ⟨i_in⟩ = Vin(t)·Ton²/(2·Lp·Ts), which is again proportional to Vin(t). The converter therefore emulates a resistance of value Re = 2·Lp·Ts/Ton² ≈ 153 Ω without any current regulator at all, and the only quantity the processor must compute is a slowly varying on-time.

The simulation confirms that the law behaves as intended. Over the two-line-cycle analysis window the on-time command TON_CMD averages 3.4309 µs with a minimum of 3.4296 µs and a maximum of 3.4322 µs – a total variation of 0.08 %. The on-time is therefore constant to within a tenth of a percent across the whole line cycle, exactly as the design requires, while the peak current varies continuously with the instantaneous line voltage. The maximum sensed peak current at the nominal 9 W load is 1.95 A, comfortably below the 2.6 A protection threshold.

8.5 Model validation and loss budget

The LTspice models carry their own measurement directives, which compute the key quantities directly from the simulated waveforms over the window 0.46 s to 0.50 s. For the PFC model at nominal resistive load these report an input RMS voltage of 35.3553 V, an input RMS current of 0.277474 A, 9.78626 W of real input power against 9.81019 VA of apparent power – a power factor of 0.997561 – an average output voltage of 11.9990 V within a band of 11.4181 V to 12.7359 V, 8.33023 W of output power and an efficiency of 85.12 %. Two independent output-power calculations (from the load resistance and from the measured load current) agree to eleven significant figures, and the output voltage at the end of the window differs from its value at the start by 1.33 mV, which confirms that the converter is in true steady state and that the averages are unbiased.

These values agree with the independently post-processed results reported in Section 10.3, which validates the analysis chain end to end: the same converter, measured inside the simulator and measured again by the external harmonic analysis, gives the same answer.

The models also report the loss contributions of the input network, which explains part of the gap between the 85 % measured efficiency and the ideal: 39.0 mW dissipated in the 0.14 Ω current-sense resistor, 23.1 mW in the winding resistance of the EMI inductor Lf, and 93.9 mW in the ESR of the X-capacitor Cx, whose RMS

current of 0.433 A is dominated by the switching-frequency component. The sense resistor and the X-capacitor together account for roughly 0.13 W, i.e. about 1.3 % of the input power – a reminder that in a 9 W converter the measurement and filtering components are themselves a significant part of the loss budget, and that the sense-resistor value is a direct trade-off between measurement resolution and efficiency. The high RMS currents implied by discontinuous conduction (Section 8.6) are what make these contributions non-negligible.

It is worth noting the difference in computational cost between the two models: the PFC model completes the simulated interval in 344 s, whereas the non-PFC model required 54,870 s – more than fifteen hours – because the narrow, high-amplitude rectifier current pulses force the solver to take very small time steps. The difficulty the simulator has with the uncorrected waveform is itself an illustration of how violent that current is.

8.6 Circuit realisation of the two models

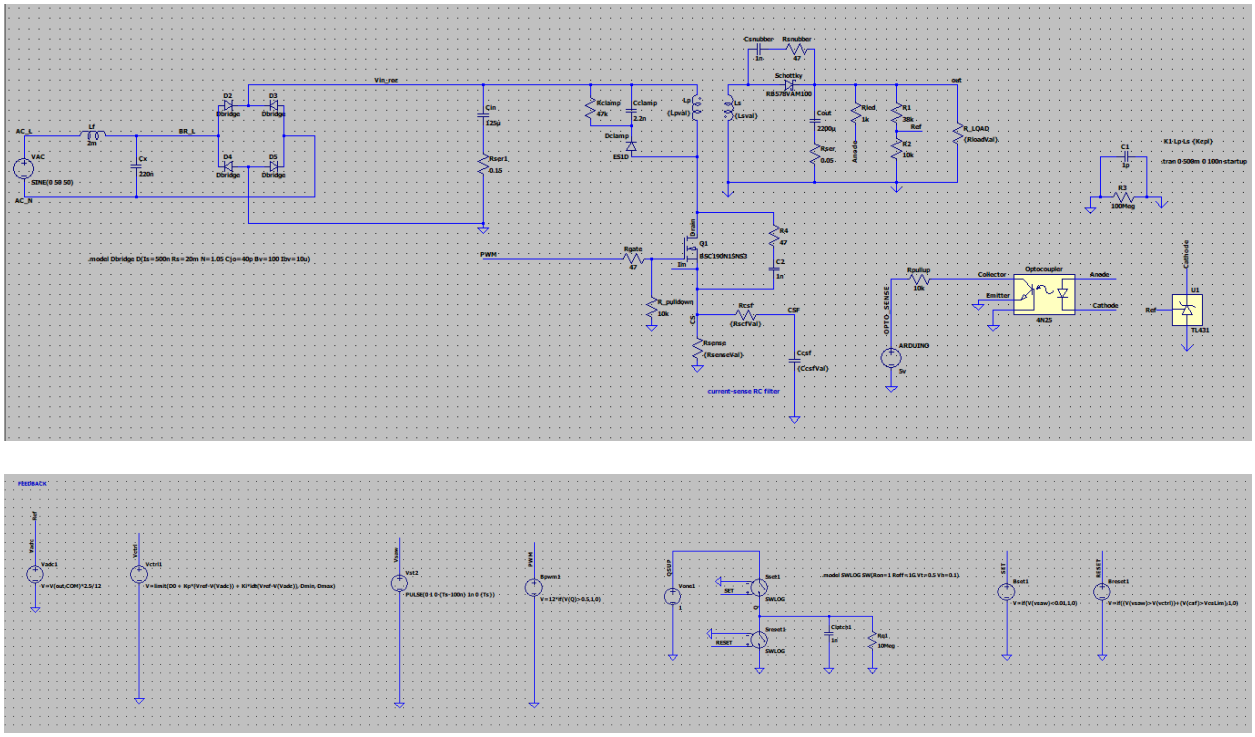


Figure 4 – LTspice schematic of our converter without power factor correction (Flyback_12v_9W_AC_corrected): power stage, isolated TL431/4N25 feedback and the behavioural model of the microcontroller duty-cycle regulator.

Figure 4 is the complete conventional model. Following the power path from the left: the mains source VAC (a 50 V peak, 50 Hz sine) feeds the differential EMI filter Lf = 2 mH and Cx = 220 nF; the diode bridge D2–D5 uses a custom model (Is = 500 nA, Rs = 20 mΩ, N = 1.05, Cjo = 40 pF, Bv = 100 V); the rectified rail Vin_rec is held by the bulk capacitor Cin = 125 µF with a series resistance Rser1 = 0.15 Ω. The primary winding Lp = 90 µH is protected by an RCD clamp (Rclamp = 47 kΩ, Cclamp = 2.2 nF and the ES1D clamp diode) that absorbs the leakage-inductance spike at turn-off. The switch Q1 is a BSC190N15NS3 MOSFET driven through Rgate = 47 Ω, with a 47 Ω / 1 nF drain snubber and a 10 kΩ pull-down that keeps the gate safe before the controller starts. The source returns to ground through Rsense = 0.14 Ω, and the sense voltage is filtered by Rcsf = 100 Ω and Ccsf = 470 pF into the node CSF.

On the secondary side, Ls = 12.7 µH is coupled to the primary with k = 0.98 – a turns ratio of about 2.66 – and feeds the RB578VAM100 Schottky rectifier, damped by a 1 nF / 47 Ω snubber, into Cout = 2200 µF with 0.05 Ω of ESR. The load R_LOAD = 16 Ω draws approximately 9 W at 12 V. The regulation path is the isolated one described in Section 8.2: the divider R1 = 38 kΩ / R2 = 10 kΩ presents the scaled output to the TL431, which

sinks current through Rled = 1 kΩ and the LED of the 4N25; the phototransistor pulls the OPTO_SENSE node against Rpullup = 10 kΩ tied to the Arduino 5 V supply.

The lower part of the sheet is the behavioural model of the firmware, and it is worth reading as pseudo-code rather than as a circuit. Vadc = V(out,COM)·2.5/12 represents the scaled ADC reading of the output voltage. Vctrl implements the PI duty regulator, limit(D0 + Kp·(Vref − Vadc) + Ki·∫(Vref − Vadc) dt, Dmin, Dmax), with D0 = 0.3, Kp = 3, Ki = 0.1, Vref = 2.5 V and the duty clamped between 0.001 and 0.45. Vsaw is a sawtooth of period Ts = 1/Fs = 10 µs, i.e. the fixed 100 kHz timer of the microcontroller. The sources Bset and Breset drive an SR latch built from two SWLOG switches: the latch is set at the start of every timer period (Vsaw < 0.01) and reset either when the sawtooth passes the duty command or when the filtered current sense exceeds VcsLim = Rsense · Ilim = 0.14 Ω × 1.6 A = 0.224 V. The latch output is buffered into the 12 V gate drive. This is a textbook fixed-frequency, peak-limited voltage-mode regulator: it controls the output perfectly and does nothing whatsoever to the shape of the input current.

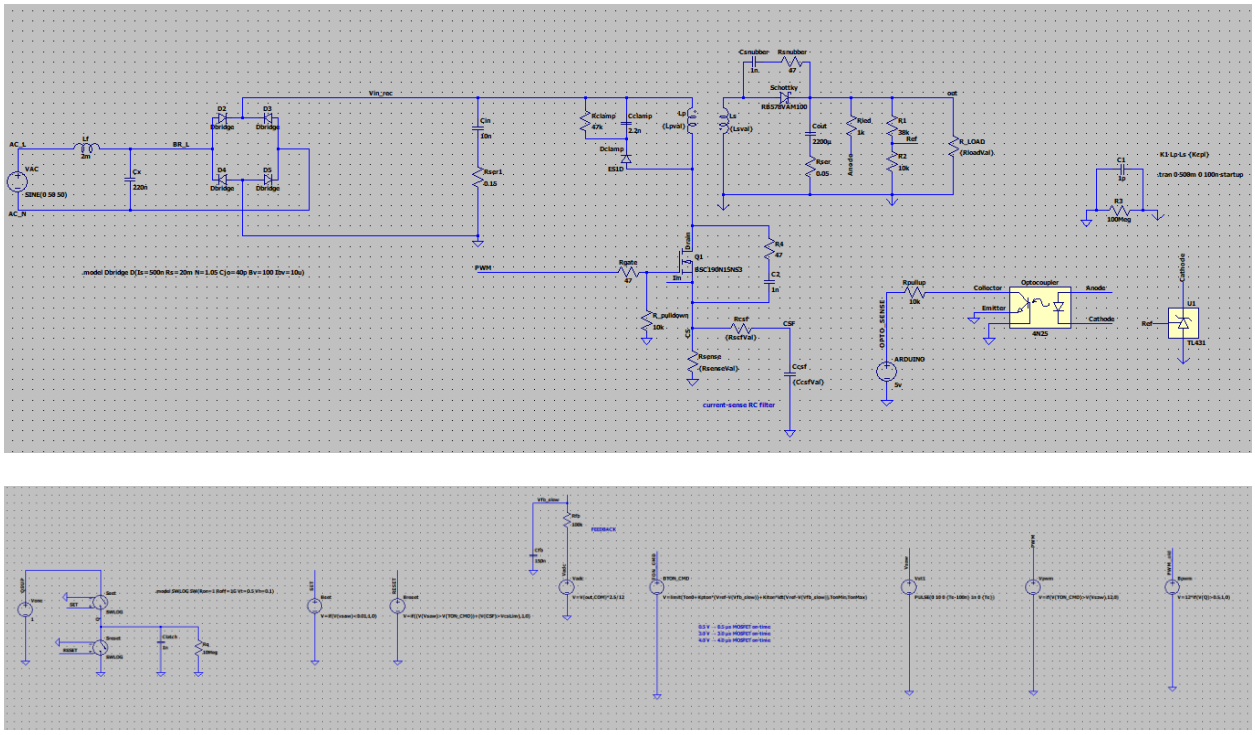


Figure 5 – LTspice schematic of our power-factor-corrected converter (Flyback_12v_9W_AC_PFC_take3): identical power stage, bulk capacitor removed, and the microcontroller model rewritten as a slow loop driving a constant on-time.

Figure 5 is the corrected model, and comparing it with Figure 4 shows how few components actually change. The power stage is identical down to the part numbers – same EMI filter, same bridge, same 90 µH / 12.7 µH transformer (here with k = 0.995), same RCD clamp, same BSC190N15NS3, same 0.14 Ω sense resistor and R–C sense filter, same Schottky and 2200 µF output capacitor, same TL431/4N25 isolated feedback. The load is R_LOAD = 17.3 Ω, again approximately 9 W.

Two changes are visible in the power path and two in the control block. In the power path, Cin drops from 125 µF to 10 nF – the bulk capacitor is gone, and what remains is only a small high-frequency decoupling capacitor, so the rail now tracks the rectified sine and the bridge conducts continuously. In the control block, the optocoupler signal is first low-pass filtered by Rfb = 100 kΩ and Cfb = 150 nF into Vfb_slow, whose 10.6 Hz corner is the deliberately slow voltage loop; and the regulator output is no longer a duty ratio but an on-time, TON_CMD = limit(Ton0 + Kpton·(Vref − Vfb_slow) + Kiton·∫(Vref − Vfb_slow) dt, TonMin, TonMax) with Ton0 = 3.0, Kpton = 0.1, Kiton = 20 and limits of 0.15 and 4.2. The sawtooth is rescaled to rise from 0 V to 10 V over the same 10 µs period, so that one volt of command corresponds to exactly one microsecond of

conduction – the annotation on the sheet states this mapping explicitly (0.5 V → 0.5 µs, 3.0 V → 3.0 µs, 4.0 V → 4.0 µs). The SR latch is unchanged: set by the 100 kHz clock, reset when the ramp passes TON_CMD or when the current sense reaches VcsLim = 0.14 Ω × 2.6 A = 0.364 V.

The essential point of the comparison is that power factor correction was obtained here by a change of control law and the removal of one capacitor, not by a redesign of the magnetics or by adding a dedicated controller chip.

8.7 Why our converter operates in DCM and not in BCM

The commercial controller studied in Chapter 7 enforces boundary conduction mode by means of a zero-current detector: an auxiliary winding tells the controller the exact instant the transformer has demagnetised, and the controller turns the switch on again immediately, so the switching frequency varies continuously along the line cycle. Our design has no auxiliary winding and no zero-current input. Turn-on is issued by the microcontroller's timer, which runs at a fixed 100 kHz; the controller sets the latch at the start of every 10 µs frame regardless of the state of the transformer. The conduction regime is therefore not chosen by the controller – it is the consequence of whether the transformer manages to demagnetise before the next clock edge.

The numbers show that it always does. With Lp = 90 µH and Ls = 12.7 µH the turns ratio is $n = \sqrt{(Lp/Ls)} \approx 2.66$, so the voltage reflected back to the primary during the off-time is $n \cdot (Vout + Vf) \approx 2.66 \times 12.7\ V \approx 34\ V$. The peak primary current at an instantaneous line voltage Vin is $Ipk = Vin \cdot Ton/Lp$, and the demagnetisation time is $tdemag = Lp \cdot Ipk / [n(Vout+Vf)] = Vin \cdot Ton / [n(Vout+Vf)]$. At the very crest of the line, Vin = 50 V and Ton = 3.43 µs give tdemag ≈ 5.1 µs, so Ton + tdemag ≈ 8.5 µs against a period of 10 µs – about 1.5 µs of dead time remain even at the worst point of the cycle. Everywhere else the instantaneous voltage is lower: at Vin = 25 V the demagnetisation takes only 2.5 µs and nearly 4 µs of dead time remain. The converter is therefore in discontinuous conduction over the entire line cycle, approaching the boundary only near the crest – which is exactly what the switching detail of Figure 7 shows, with the secondary current reaching zero and the drain voltage ringing freely before the next fixed-frequency turn-on.

This was a design decision, not an oversight, and it rests on the observation that power factor correction does not require BCM. What the mains sees is the cycle-averaged input current, and in discontinuous conduction with a fixed frequency that average is $\langle i_in \rangle = Vin(t) \cdot Ton^2/(2 \cdot Lp \cdot Ts)$: with Ton and Ts both constant over the line cycle it is exactly proportional to the instantaneous line voltage, which is the definition of resistor emulation. In BCM the same proportionality is obtained with Ts allowed to vary. Both laws give a high power factor; DCM obtains it at a fixed frequency. Substituting the simulated values gives an emulated resistance $Re = 2 \cdot Lp \cdot Ts/Ton^2 \approx 153\ \Omega$ and an input power of about 8 W, against the 9.79 W actually measured – the difference being the leakage-inductance and clamp losses and the imperfect current build-up near the zero crossings, which the ideal expression neglects.

The practical reasons for preferring DCM in a microcontroller-based design are decisive at this power level:

- No auxiliary winding and no zero-current interrupt. BCM would require the processor to respond to a demagnetisation event within tens of nanoseconds, up to several hundred thousand times per second. That is an analogue comparator's job, not an Arduino's. In DCM the processor only has to program one timer compare register with a slowly varying number.
- Fixed frequency. The switching frequency is set by the timer and never moves, so the EMI spectrum is concentrated at 100 kHz and its harmonics and the input filter can be designed for it – whereas the variable frequency of BCM spreads the noise over a wide band and complicates the filter, as noted in Section 5.3.
- Deterministic timing. Every switching cycle is identical in duration, which makes the firmware trivially predictable: the control update, the ADC sampling and the PWM update all run on the same 10 µs grid.

- No reverse recovery. Like BCM, DCM turns the output diode off at zero current, so the reverse-recovery losses of CCM are avoided.
- The price paid. Because the current must fall to zero and wait, the peak and RMS currents are higher than they would be in BCM for the same output power – 1.95 A of peak primary current for 0.7 A of output current – which raises conduction losses in the switch, the sense resistor and the X-capacitor, and the switch turns on into a partially resonant drain voltage rather than at zero current, so the turn-on loss that BCM eliminates is still present. At 9 W these costs are small; at several hundred watts they would not be, and BCM or CCM would become the correct choice.

8.8 Why an Arduino-class microcontroller was chosen as the controller

The choice of an Arduino-class microcontroller in place of the dedicated MP44010 device was deliberate, and it was driven by three considerations: cost, convenience of development, and the didactic value of the resulting platform. It is worth stating them explicitly, because they also explain several of the design decisions already described in this chapter.

- Cost. An Arduino-class board (an ATmega328P or an equivalent 8-bit device) costs a few units of currency, is stocked by every electronics distributor and is second-sourced many times over, whereas a dedicated PFC controller is a single-source part with a fixed function. In a 9 W converter the controller is a visible fraction of the bill of materials, and the digital implementation also deletes external parts: no multiplier network on the MULT pin, no auxiliary winding and no zero-current-detect components, since the on-time is issued by a timer rather than by a demagnetisation event (Section 8.7). The saving is therefore not only the price of the chip but the component count around it.
- Convenience. The control law lives in firmware, so changing it costs a recompilation rather than a new board: Ton0, Kp, Ki, the clamps TonMin and TonMax and even the choice between the duty-cycle law of Section 8.3 and the constant-on-time law of Section 8.4 are all edits to a few lines of code on an otherwise identical power stage. That is exactly what made the two-variant comparison of this report possible. The same processor also gives, for free, facilities an analogue controller does not have: serial logging of Vout, Ton and the measured error while the converter runs, soft-start and protection thresholds expressed as numbers, and the possibility of adding supervision or communication later without touching the hardware. The toolchain is free and familiar, which shortens the development loop considerably.
- Suitability as a teaching tool. This is a laboratory project, and a platform whose control loop is readable source code is worth a great deal pedagogically. With a dedicated controller the loop is sealed inside the package and the student can only observe its effects; here every element of the theory of Chapters 2 and 7 appears as an explicit line of code – the reference, the error, the integrator, the limit function and the timer compare register that becomes the on-time. A student can deliberately break the design and watch the physics respond: shortening the feedback filter time constant until the loop starts to track within the line half-cycle and the power factor collapses, changing Ton0 and watching the emulated resistance $Re = 2 \cdot Lp \cdot Ts/Ton^2$ move, or switching between the corrected and uncorrected firmware and measuring the difference at the input terminals. The laboratory session reproduced in Appendix A is built around precisely this kind of experiment.

The price of the choice is the one already quantified in Section 8.7: an 8-bit processor cannot close a fast current loop or service a zero-current interrupt at 100 kHz, which is why the design uses fixed-frequency constant-on-time control in DCM rather than the BCM scheme of the reference unit, and accepts the higher peak and RMS currents that follow. At 9 W that is an acceptable trade; the point of the comparison with the MP44010 is to show exactly what is given up and what is gained.

9. Simulation Environment and Measurement Method

Before the laboratory measurements, our two converter models of Chapter 8 – diode bridge, Flyback power stage, TL431/4N25 isolated feedback, microcontroller control law and output loads – were simulated at the transistor level in LTspice, and the resulting waveforms were post-processed numerically in order to extract the power-quality metrics. The simulation therefore serves two purposes: it validates the theoretical model of Chapters 5–8 and the control law we chose, and it provides the quantitative prediction against which the laboratory results will later be compared.

All the results reported in Chapter 10 were obtained under a single, common set of conditions, so that the different cases can be compared directly:

- Input source: a pure sinusoidal AC source of 35.3553 V RMS (50 V peak) at the line frequency; the measured input-voltage THD is 0.000 %, so any distortion observed in the results is created by the converter alone and not by the source.
- Output: a regulated 12 V DC rail feeding the load under test.
- Analysis window: 0.460000 s to 0.500000 s – exactly two complete line cycles, taken well after the start-up transient has decayed, so that the converter is in steady state and the averages are unbiased.
- Extracted metrics: input voltage and current RMS, average input power, apparent power, true power factor, fundamental displacement factor, V–I fundamental phase difference, current distortion factor, input-current THD (harmonics 2–40 and residual), average output voltage, current and power, efficiency, and the drift of the output voltage across the window.
- Reported figures per case: an input voltage/current waveform plot, an input-current harmonic spectrum normalised to the fundamental, a numerical summary of the analysis window, and a table of the first 20 harmonics of voltage and current.

Two families of experiments were run on our design. The first family (Sections 10.1–10.5) compares our non-PFC model with our PFC model, and then holds the output power essentially constant while changing only the character of the load – resistive, inductive, capacitive – which is the direct test of the research hypothesis. The second family (Section 10.6) keeps the load character fixed and changes only the magnitude and dynamics of the load step, and therefore isolates the effect of the power level from the effect of the load type.

10. Simulation Results

10.1 Verification of the switching regime and of the control law

The first simulations verify the switching behaviour of the two models at the cycle level, before any power-quality metric is computed. Three traces are shown in each figure: the gate command V(pwm), the secondary (Schottky) diode current I(Schottky), and the drain-to-source voltage V(drain,CS). Both models run at the same fixed 100 kHz timer frequency and both operate in discontinuous conduction mode, for the reasons given in Section 8.7; what these figures establish is that the switching cycle behaves as designed and that the difference between the two models lies in the control law and the input capacitance, not in the power stage.

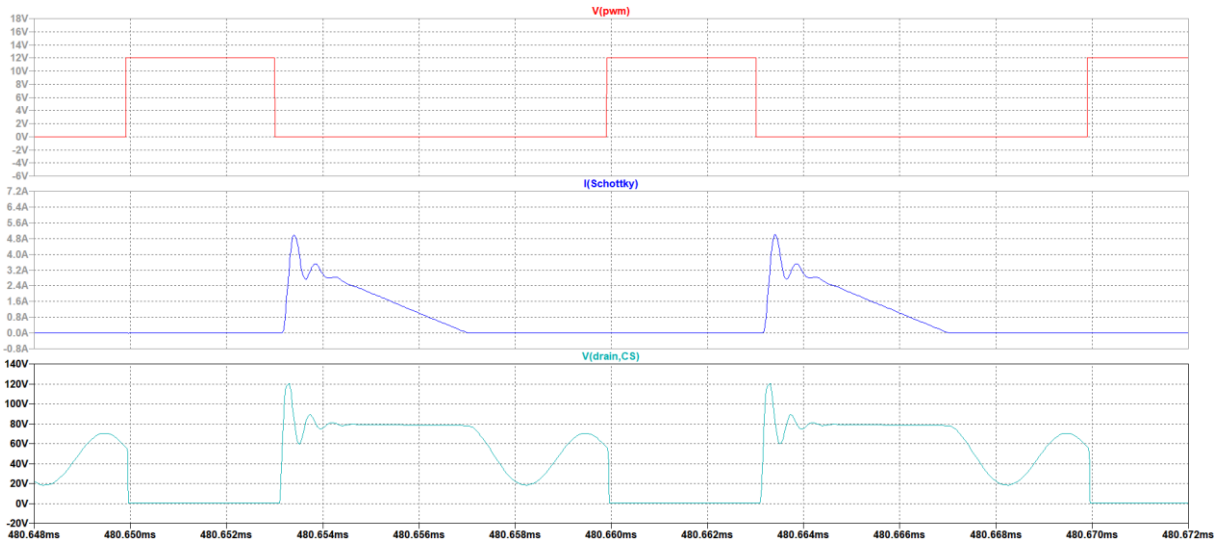


Figure 6 – Switching detail of our non-PFC model: gate signal, secondary diode current and drain voltage over two switching cycles, showing the dead time and free ringing of DCM.

Figure 6 shows our non-PFC model (Flyback_12v_9W_AC_corrected). The switching period is 10 µs, as set by the timer. The secondary current falls to zero at about 480.657 ms while the next gate pulse arrives only at about 480.662 ms; between these instants the current stays flat at zero and the drain voltage rings freely – the resonant oscillation between the primary inductance and the drain capacitance, clearly visible as the damped sinusoid between 80 V and 20 V. This dead time is the signature of discontinuous conduction: the switch is turned on at an arbitrary point of the ringing, so the turn-on is hard. Note the peak diode current of about 5 A: because the bulk capacitor restricts conduction to the crest of the line, all the energy of the half-cycle must be transferred in a few switching cycles, and the peaks are correspondingly violent.

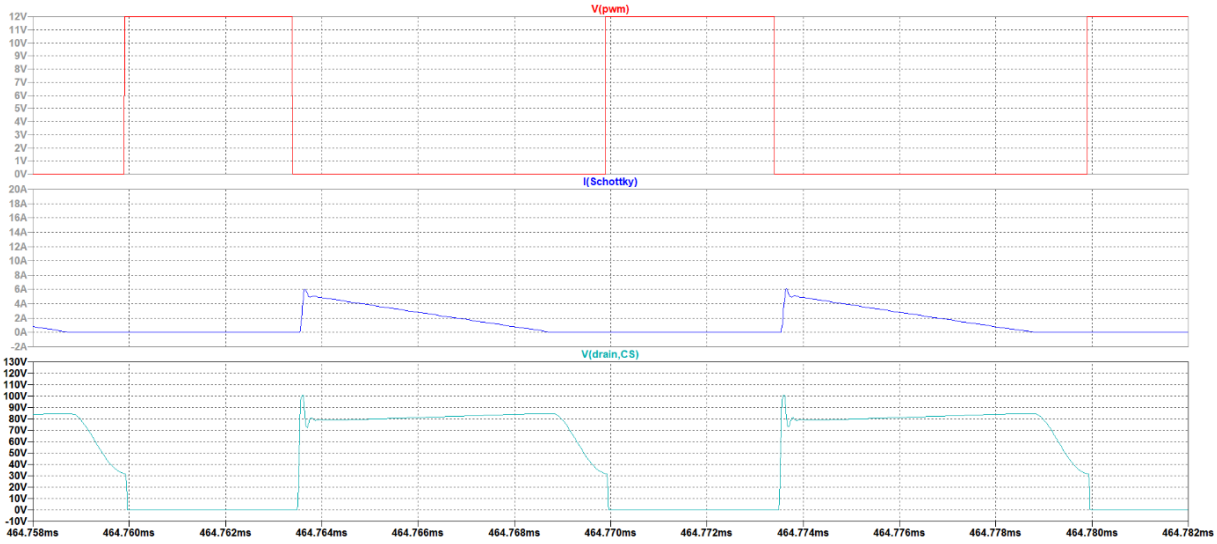


Figure 7 – Switching detail of our PFC model near the line crest: constant 3.43 µs on-time, linear demagnetisation and fixed-frequency turn-on (DCM).

Figure 7 shows our PFC model (Flyback_12v_9W_AC_PFC_take3) on a resistive load, captured near the crest of the line where the converter works hardest. The period is again 10 µs – the timer frequency is unchanged – but the on-pulse now has the constant 3.43 µs length set by TON_CMD, and the secondary current is a clean triangle that falls linearly from about 4.7 A to zero over roughly 5 µs, in agreement with the demagnetisation time computed in Section 8.7. The remaining interval before the next clock edge is short here, so the converter is close to the conduction boundary at this point of the line cycle; away from the crest the dead time grows and

the converter is firmly discontinuous. The drain voltage confirms the sequence: the reflected plateau of about 80 V during demagnetisation, then the decay towards the input rail once the transformer is empty, then the next fixed-frequency turn-on. Comparison of the two figures shows the same power stage doing the same kind of switching under two different control laws – and it is the law, not the regime, that produces the difference in power factor.

10.2 Baseline reference – converter without PFC

To quantify the benefit of the PFC control law, our converter was first simulated in its non-PFC form (Section 8.3), feeding an equivalent output power. This case is the reference against which every subsequent result is judged.

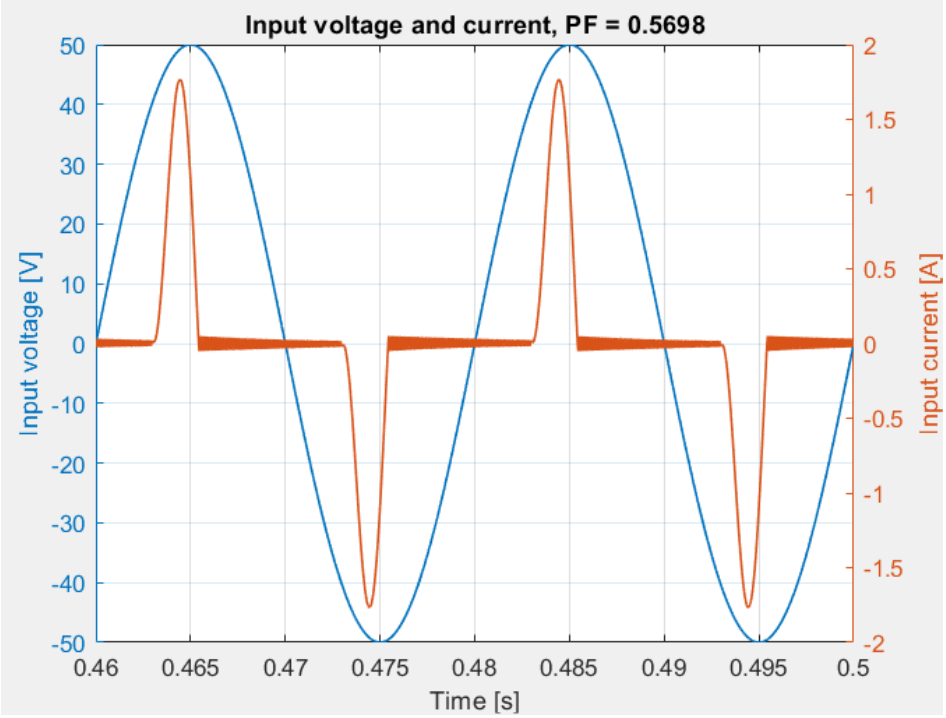


Figure 8 – Input voltage and input current without PFC; PF = 0.5698.

Figure 8 shows the classic signature of an uncorrected rectifier with a bulk capacitor. The input current is not a sine wave at all: it is zero over most of the line cycle and flows only in a narrow pulse of about 1.8 A around the crest of the voltage, when the instantaneous line voltage exceeds the capacitor voltage. The current pulse is roughly aligned with the voltage peak – so the displacement is small – but the waveform is extremely distorted, and the true power factor collapses to 0.5698.

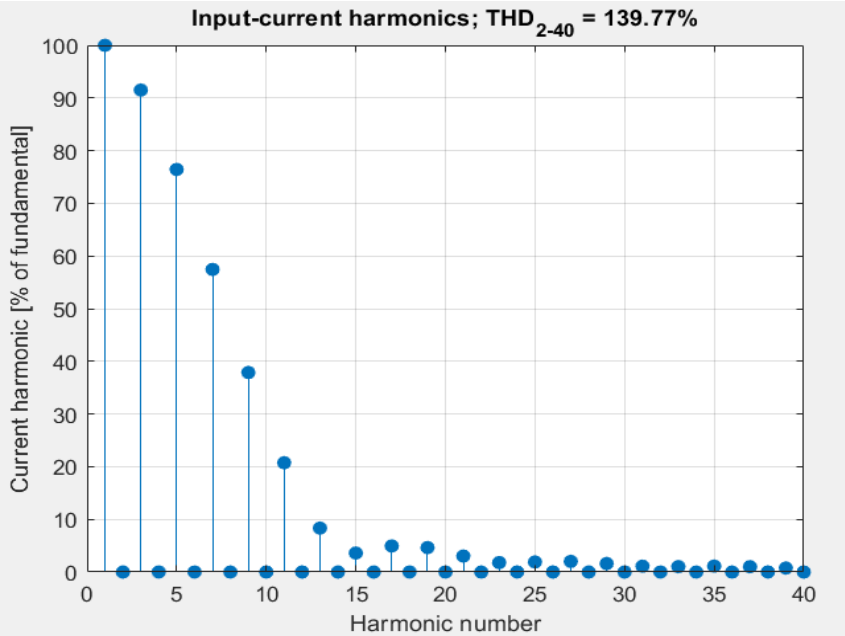


Figure 9 – Input-current harmonic spectrum without PFC; THD(2–40) = 139.77 %.

Figure 9 explains where that power factor is lost. The spectrum is dominated by odd harmonics of very large amplitude: the third harmonic reaches 91.5 % of the fundamental, the fifth 76.4 %, the seventh 57.4 %, the ninth 37.9 % and the eleventh 20.7 %; the even harmonics are essentially zero, as expected for a half-wave-symmetric waveform. The resulting total harmonic distortion is 139.77 % – the harmonic content of the current is larger than the fundamental itself. A converter with this spectrum would fail IEC 1000-3-2 by a wide margin.

```
Analysis window: 0.460000 s to 0.500000 s (2.000 line cycles)
-------------------------------------------------------------
Input voltage RMS                 =      35.3553 V
Input current RMS                 =     0.562314 A
Average input power               =      11.3287 W
Input apparent power              =      19.8808 VA
TRUE power factor                 =     0.569830
Fundamental displacement PF       =     0.980041
V-I fundamental phase difference  =   -11.466664 deg
Current distortion factor I1/Irms =     0.581435

Input-current THD, harmonics 2-40 =   139.765967 %
Input-current residual THD        =   139.928338 %
Input-voltage THD, harmonics 2-40 =     0.000000 %
Input-voltage residual THD        =     0.000000 %

Average output voltage            =      12.0512 V
Average output current            =       0.7532 A
Average output power              =      9.07728 W
EFFICIENCY                        =    80.126576 %
Output-voltage drift over window  =     -0.00011 V (-0.000912772 %)
```

Figure 10 – Numerical summary of the analysis window, without PFC.

Figure 10 gives the numerical proof and, importantly, separates the two mechanisms of power-factor loss. The fundamental displacement factor is 0.9800 and the V–I fundamental phase difference is only −11.47°, so displacement accounts for almost nothing; the distortion factor, however, is 0.5814, and it is this term that drags the true power factor down to 0.5698. The input draws 0.5623 A RMS and 19.88 VA of apparent power to deliver only 11.33 W of real power, of which 9.08 W reaches the load – an efficiency of 80.13 %. The output voltage is well regulated at 12.05 V with a drift of −0.0009 % across the window, which confirms that the poor power factor is an input-side phenomenon and not a regulation failure.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.32695	100
2	1.2738e-09	3.6027e-09	4.3419e-06	0.001328
3	3.1453e-09	8.8963e-09	0.29916	91.501
4	1.2083e-09	3.4177e-09	5.1998e-06	0.0015904
5	5.6057e-09	1.5855e-08	0.24986	76.423
6	3.8945e-09	1.1015e-08	5.484e-06	0.0016773
7	2.2793e-09	6.447e-09	0.18782	57.446
8	1.7874e-09	5.0556e-09	4.9131e-06	0.0015027
9	6.7819e-09	1.9182e-08	0.12385	37.88
10	4.7141e-09	1.3333e-08	3.6039e-06	0.0011023
11	2.6344e-09	7.4512e-09	0.067834	20.748
12	4.0184e-09	1.1366e-08	2.0608e-06	0.0006303
13	4.7443e-09	1.3419e-08	0.027254	8.3359
14	1.683e-09	4.7601e-09	8.675e-07	0.00026533
15	9.654e-09	2.7306e-08	0.011787	3.6051
16	1.7464e-09	4.9395e-09	8.6974e-07	0.00026602
17	5.6284e-09	1.5919e-08	0.016152	4.9403
18	6.1302e-09	1.7339e-08	1.006e-06	0.0003077
19	8.185e-09	2.3151e-08	0.015171	4.64
20	4.9351e-09	1.3958e-08	7.7717e-07	0.0002377

Figure 11 – First 20 harmonics of the input voltage and input current, without PFC.

Figure 11 lists the same spectrum in numerical form. The voltage harmonics are all of order 10^{-9} V, i.e. the source is perfectly clean, so every current harmonic in the table is generated by the converter. The current fundamental is 0.32695 A while the total RMS current is 0.5623 A – the difference is carried entirely by the harmonics listed (0.29916 A at the third, 0.24986 A at the fifth, 0.18782 A at the seventh), which is the tabular expression of the 0.5814 distortion factor.

10.3 PFC active – resistive load (baseline with correction)

With our digital constant-on-time control law active (Section 8.4) and a purely resistive output load, the same four figures were produced. This case is the reference point for the load-character comparison of Sections 10.4 and 10.5.

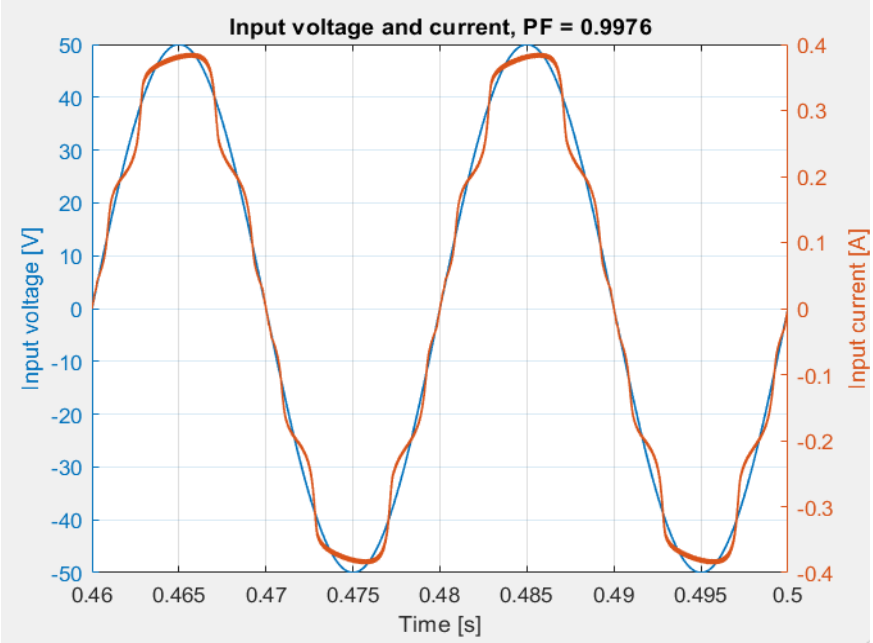


Figure 12 – Input voltage and current, PFC active, resistive load; PF = 0.9976.

Figure 12 should be compared directly with Figure 8. The narrow current pulse has disappeared: the current now follows the voltage over the whole cycle, reaching about 0.39 A at the crest, and it crosses zero together with the voltage. The residual features are the fine high-frequency ripple of the DCM switching (the thickness of the trace) and a slight flattening around the crest. The true power factor is 0.9976 – a 75 % improvement over the uncorrected case – while the peak current has fallen by a factor of roughly 4.6 for comparable output power.

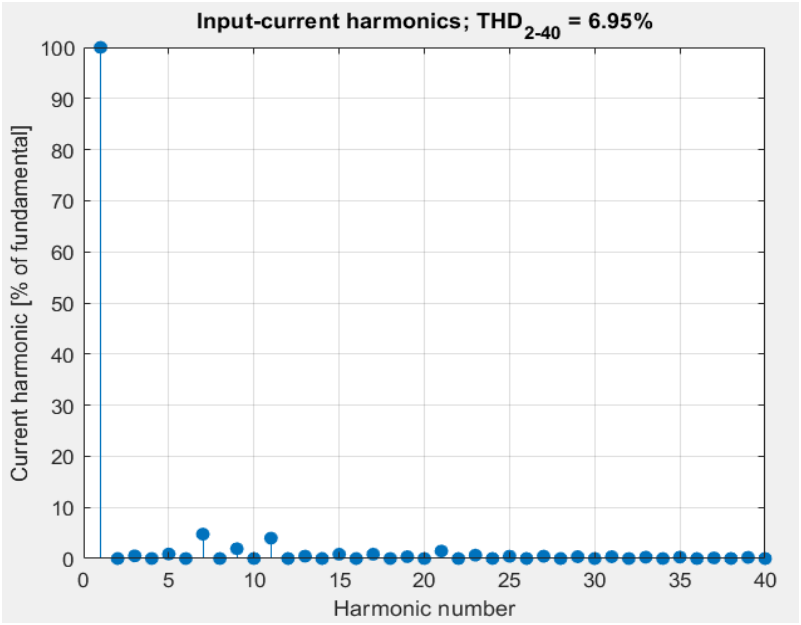


Figure 13 – Input-current harmonic spectrum, resistive load; THD(2–40) = 6.95 %.

Figure 13 shows the collapse of the harmonic content. The low-order harmonics that dominated Figure 9 are gone – the third is 0.52 % and the fifth 0.88 % – and the largest remaining components are the seventh at 4.77 % and the eleventh at 3.98 %, which are the residue of the zero-crossing distortion discussed in Section 7.3 and of the small flattening near the crest. The overall THD is 6.95 %, a twenty-fold reduction from 139.77 %.

```
Analysis window: 0.460000 s to 0.500000 s (2.000 line cycles)
------------------------------------------------------------
Input voltage RMS                 =      35.3553 V
Input current RMS                 =     0.277473 A
Average input power               =      9.78623 W
Input apparent power              =      9.81015 VA
TRUE power factor                 =     0.997562
Fundamental displacement PF       =     1.000000
V-I fundamental phase difference  =     0.010828 deg
Current distortion factor I1/Irms =     0.997562

Input-current THD, harmonics 2-40 =     6.953314 %
Input-current residual THD        =      6.996311 %
Input-voltage THD, harmonics 2-40 =     0.000000 %
Input-voltage residual THD        =      0.000254 %

Average output voltage            =        11.999 V
Average output current            =     0.693586 A
Average output power              =      8.33025 W
EFFICIENCY                        =    85.122188 %
Output-voltage drift over window  =      0.00133 V (0.0110842 %)
```

Figure 14 – Numerical summary of the analysis window, resistive load.

Figure 14 confirms the mechanism. The displacement factor is 1.000000 with a phase difference of only 0.011°, and the distortion factor is 0.9976 – both terms of the power factor are now essentially ideal. The input draws 0.2775 A RMS and 9.81 VA to deliver 9.79 W of real power, so almost all of the apparent power is useful; 8.33 W reaches the load, an efficiency of 85.12 %, five points better than the uncorrected case because the much lower RMS current reduces the conduction losses. The output sits at 11.999 V with a drift of 0.011 %.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.2768	100
2	3.9349e-09	1.1129e-08	2.5328e-06	0.00091503
3	5.4324e-09	1.5365e-08	0.0014315	0.51716
4	2.6598e-09	7.5229e-09	1.7439e-06	0.00063003
5	3.2838e-09	9.2881e-09	0.0024251	0.87613
6	2.7477e-09	7.7716e-09	1.7497e-06	0.00063212
7	8.9735e-09	2.5381e-08	0.013191	4.7655
8	4.9488e-09	1.3997e-08	7.6953e-07	0.00027801
9	2.6972e-09	7.6288e-09	0.005289	1.9108
10	2.0587e-09	5.8228e-09	2.1045e-06	0.00076029
11	7.8603e-09	2.2232e-08	0.011007	3.9764
12	4.1584e-09	1.1762e-08	1.5224e-06	0.00055
13	7.7462e-09	2.191e-08	0.00127	0.45883
14	8.2742e-09	2.3403e-08	1.4279e-06	0.00051585
15	3.8372e-09	1.0853e-08	0.0023335	0.84303
16	6.5129e-09	1.8421e-08	1.0885e-06	0.00039325
17	4.5818e-09	1.2959e-08	0.002318	0.83745
18	1.2709e-09	3.5947e-09	7.2692e-07	0.00026262
19	4.2259e-09	1.1953e-08	0.00096896	0.35006
20	2.2586e-09	6.3884e-09	2.1919e-06	0.0007919

Figure 15 – First 20 harmonics, resistive load.

Figure 15 quantifies the residual spectrum: the fundamental is 0.2768 A out of a total 0.2775 A RMS, and the largest harmonic currents are 13.2 mA at the seventh and 11.0 mA at the eleventh – three orders of magnitude below the fundamental. This table is the baseline against which the inductive and capacitive cases are compared.

10.4 PFC active – inductive load

The output resistor was replaced by an inductive load at the same power level, leaving everything else unchanged. If the research hypothesis is correct, the input-side results must be indistinguishable from Section 10.3.

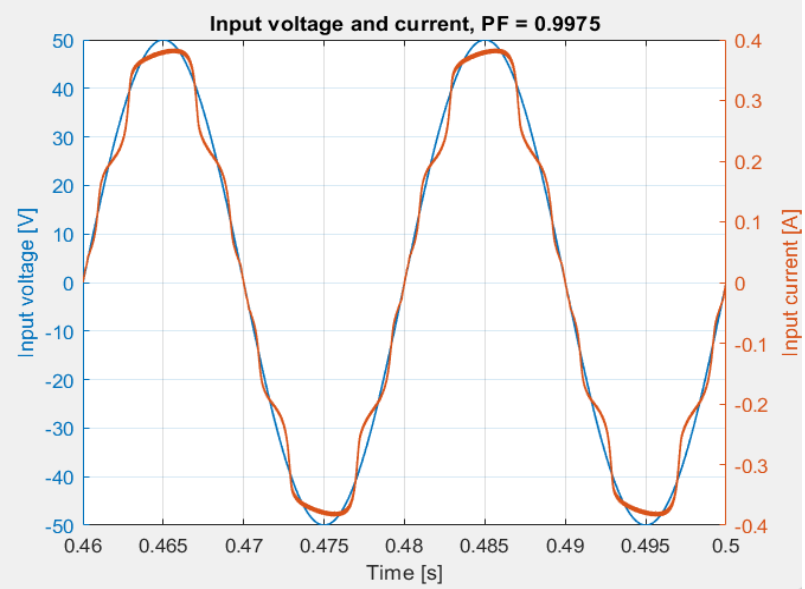


Figure 16 – Input voltage and current, inductive load; PF = 0.9975.

Figure 16 is, to the eye, the same waveform as Figure 12: the same sinusoidal current of about 0.39 A peak, in phase with the voltage, with the same slight crest flattening. The inductance at the output produces no visible phase lag at the input, because the input current is not determined by the load impedance at all – it is synthesised by the multiplier from the rectified line voltage. The power factor is 0.9975 against 0.9976 for the resistive case, a difference of 0.01 %.

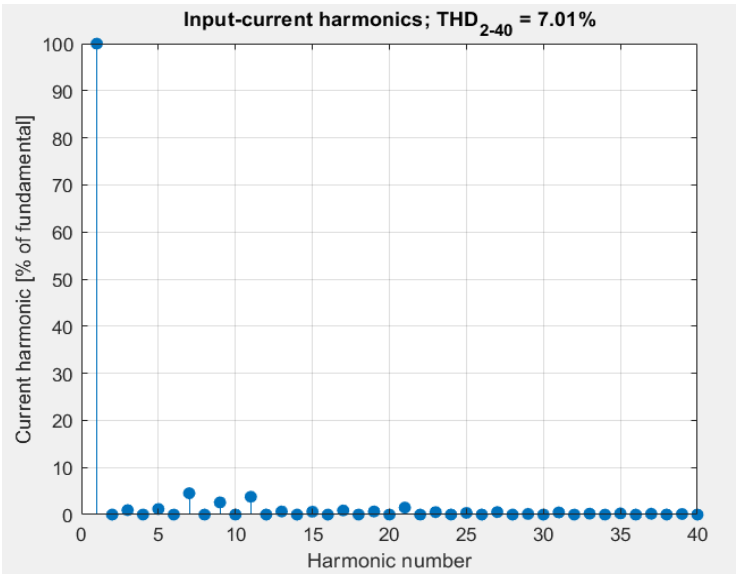


Figure 17 – Input-current harmonic spectrum, inductive load; THD(2–40) = 7.01 %.

Figure 17 reproduces the resistive spectrum almost exactly – seventh harmonic near 4.7 %, eleventh near 4 %, everything else below 3 % – and gives a THD of 7.01 % against 6.95 %. The harmonic signature therefore belongs to the converter and its control law, not to the load.

```
Analysis window: 0.460000 s to 0.500000 s (2.000 line cycles)
-------------------------------------------------------------
Input voltage RMS                 =      35.3553 V
Input current RMS                 =     0.274501 A
Average input power               =        9.681 W
Input apparent power              =      9.70506 VA
TRUE power factor                 =     0.997521
Fundamental displacement PF       =     1.000000
V-I fundamental phase difference  =    -0.017906 deg
Current distortion factor I1/Irms =     0.997521

Input-current THD, harmonics 2-40 =     7.011019 %
Input-current residual THD        =      7.055048 %
Input-voltage THD, harmonics 2-40 =     0.000000 %
Input-voltage residual THD        =      0.000254 %

Average output voltage            =       11.9994 V
Average output current            =     0.685678 A
Average output power              =       8.23521 W
EFFICIENCY                        =    85.065700 %
Output-voltage drift over window  =       0.00129 V (0.0107506 %)
```

Figure 18 – Numerical summary of the analysis window, inductive load.

Figure 18 provides the decisive numbers. The displacement factor is 1.000000 and the V–I phase difference is −0.018°: the reactive character of the load is not reflected to the mains even in the smallest degree, because the DC bus capacitors and the slow voltage loop absorb it entirely. The input takes 0.2745 A RMS and 9.71 VA for 10.68 W of real power, delivering 8.24 W at 11.9994 V – an efficiency of 85.07 %, within 0.06 points of the resistive case.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.27382	100
2	4.2841e-09	1.2117e-08	2.0789e-06	0.00075924
3	5.4306e-09	1.536e-08	0.0025829	0.94329
4	5.6566e-09	1.5999e-08	1.3522e-06	0.00049381
5	4.4165e-09	1.2492e-08	0.0032361	1.1818
6	5.8876e-09	1.6653e-08	2.1763e-06	0.00079478
7	1.6545e-09	4.6796e-09	0.01242	4.5359
8	1.7587e-09	4.9743e-09	1.1353e-06	0.00041462
9	6.627e-09	1.8744e-08	0.0070839	2.5871
10	4.1409e-09	1.1712e-08	2.4413e-06	0.00089157
11	6.5126e-09	1.842e-08	0.010367	3.7861
12	1.6421e-09	4.6446e-09	1.494e-06	0.00054562
13	4.3143e-09	1.2203e-08	0.0018026	0.65832
14	1.6975e-09	4.8014e-09	1.0104e-06	0.00036899
15	3.4963e-09	9.8889e-09	0.0016765	0.61228
16	4.4671e-09	1.2635e-08	1.2627e-06	0.00046113
17	4.9362e-09	1.3962e-08	0.0023981	0.87581
18	6.67e-09	1.8866e-08	4.6149e-07	0.00016854
19	4.5433e-09	1.285e-08	0.0018087	0.66053
20	3.8038e-09	1.0759e-08	1.2055e-06	0.00044027

Figure 19 – First 20 harmonics, inductive load.

Figure 19 confirms the numerical identity of the two spectra: the same harmonic orders dominate at the same per-unit amplitudes, scaled only by the marginally smaller fundamental. Comparing Figures 15 and 19 line by line is the strictest form of the test, and the two tables agree to within a fraction of a percent.

10.5 PFC active – capacitive load

The third load character is capacitive, again at the same power level. This is the most demanding of the three static cases, because a capacitive load draws its energy in charging bursts rather than continuously.

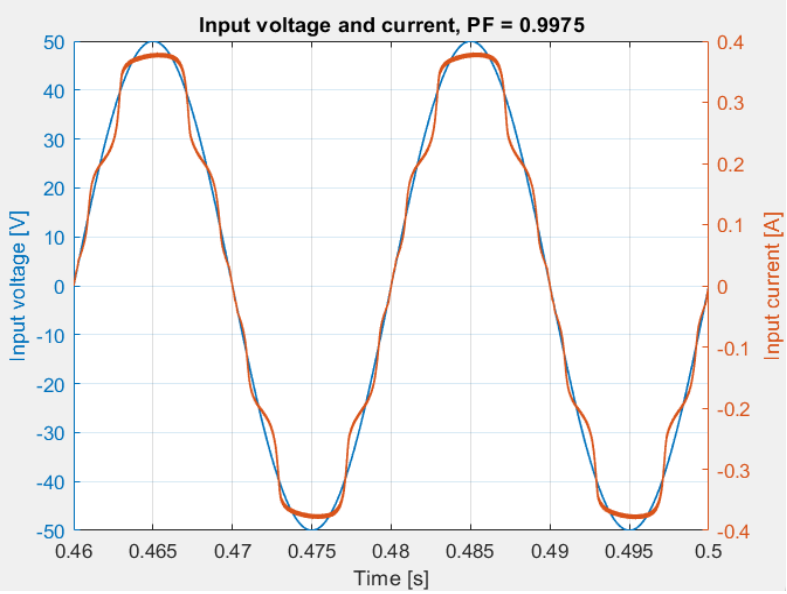


Figure 20 – Input voltage and current, capacitive load; PF = 0.9975.

Figure 20 again shows a clean sinusoidal input current of about 0.38 A peak, in phase with the voltage. The charging bursts drawn by the capacitive load on the DC side are invisible at the input: they are supplied by the bulk capacitance and averaged by the 10 Hz voltage loop, exactly as predicted in Chapter 2. The power factor is 0.9975.

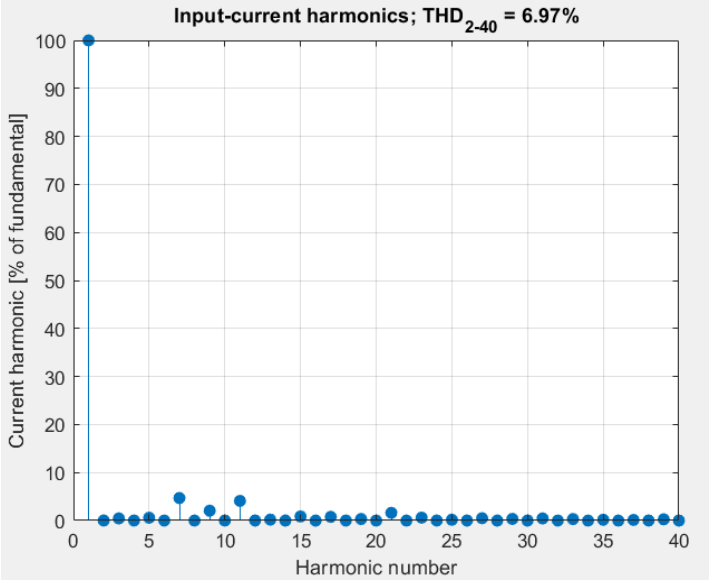


Figure 21 – Input-current harmonic spectrum, capacitive load; THD(2–40) = 6.97 %.

Figure 21 is again the same spectrum – dominant seventh at about 4.8 % and eleventh at about 4 % – with a THD of 6.97 %, which falls between the resistive (6.95 %) and inductive (7.01 %) values. The spread across the three load characters is 0.06 percentage points, far below any measurement tolerance.

```
Analysis window: 0.460000 s to 0.500000 s (2.000 line cycles)
------------------------------------------------------------
Input voltage RMS                 =      35.3553 V
Input current RMS                 =     0.275107 A
Average input power               =      9.70265 W
Input apparent power              =      9.72651 VA
TRUE power factor                 =     0.997547
Fundamental displacement PF       =     1.000000
V-I fundamental phase difference  =     0.037196 deg
Current distortion factor I1/Irms =     0.997547

Input-current THD, harmonics 2-40 =     6.972857 %
Input-current residual THD         =     7.017134 %
Input-voltage THD, harmonics 2-40 =     0.000000 %
Input-voltage residual THD         =     0.000248 %

Average output voltage             =       11.985 V
Average output current             =     0.692775 A
Average output power               =      8.30489 W
EFFICIENCY                          =    85.594023 %
Output-voltage drift over window   =      0.02055 V (0.171464 %)
```

Figure 22 – Numerical summary of the analysis window, capacitive load.

Figure 22 records a displacement factor of 1.000000, a phase difference of 0.037°, a distortion factor of 0.9975, 0.2751 A RMS input current, 9.70 W input power, 8.30 W output power and an efficiency of 85.59 %. The output-voltage drift is 0.171 % – the largest of the three static cases, which is the only visible trace of the capacitive load, and it is still an order of magnitude smaller than the ripple specification.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.27443	100
2	1.0679e-08	3.0205e-08	2.6519e-05	0.0096632
3	6.8562e-09	1.9392e-08	0.001213	0.44202
4	1.7034e-09	4.8178e-09	8.9183e-07	0.00032497
5	5.4143e-09	1.5314e-08	0.001591	0.57974
6	9.5601e-10	2.704e-09	2.5617e-06	0.00093344
7	4.1204e-09	1.1654e-08	0.012855	4.6843
8	4.1399e-09	1.1709e-08	5.0118e-06	0.0018262
9	3.8109e-09	1.0779e-08	0.0056068	2.0431
10	6.8678e-09	1.9425e-08	7.2388e-06	0.0026377
11	4.0264e-09	1.1388e-08	0.011241	4.0961
12	2.179e-09	6.1633e-09	4.1922e-06	0.0015276
13	8.5864e-10	2.4286e-09	0.00052094	0.18983
14	9.2102e-09	2.605e-08	1.5214e-06	0.00055439
15	3.7923e-09	1.0726e-08	0.0023226	0.84633
16	3.9846e-09	1.127e-08	2.5965e-06	0.00094614
17	2.7667e-09	7.8253e-09	0.002113	0.76996
18	2.9049e-09	8.2164e-09	1.5423e-06	0.000562
19	7.7608e-09	2.1951e-08	0.00089752	0.32705
20	4.0688e-09	1.1508e-08	2.5566e-06	0.00093159

Figure 23 – First 20 harmonics, capacitive load.

Figure 23 closes the set. Taken together, Figures 12–23 show three fundamentally different output impedances producing input-side results that differ by 0.01 % in power factor, 0.06 points in THD and 0.5 points in efficiency. This is the core experimental confirmation of the research hypothesis.

10.6 Time-varying load

The static tests establish independence from the character of the load. The remaining question is what happens when the load changes in time. In this family of simulations a fixed load is combined with a pulsed load that is switched on for a 5 ms interval twice per line cycle, so that the total output power steps repeatedly between two levels. Four step sizes were examined: 50 %→100 %, 100 %→150 %, 100 %→200 % and 100 %→400 % of the nominal load. In each case five figures are reported: the input waveforms, the harmonic spectrum, the numerical summary, the harmonic table, and a power plot showing the fixed load, the pulsed load and the total useful output power.

10.6.1 Step from 50 % to 100 %

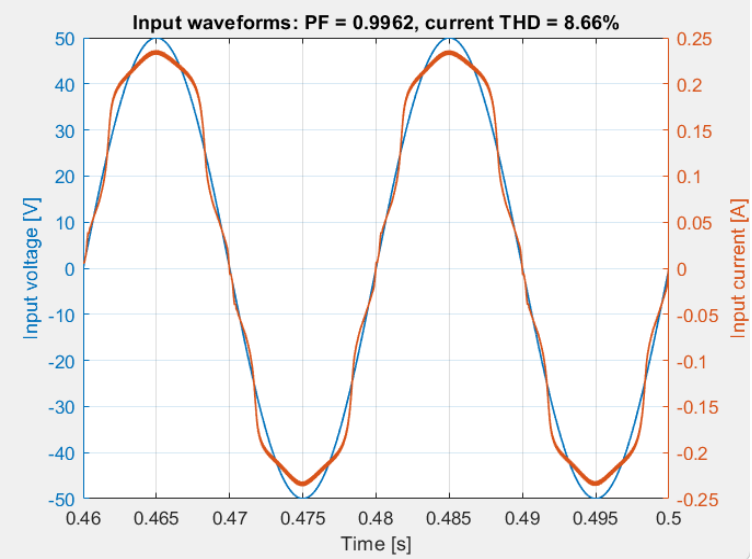


Figure 24 – Input waveforms, 50 %→100 % load step; PF = 0.9962, current THD = 8.66 %.

Figure 24 shows that the repeated doubling of the load is almost invisible at the input. The current remains sinusoidal at about 0.24 A peak and in phase with the voltage; the only sign of the pulsing is a slight asymmetry in the shape of the negative half-cycle, where the pulse happens to fall. The power factor is 0.9962 and the THD 8.66 % – a degradation of only 1.7 points of THD relative to the static resistive case.

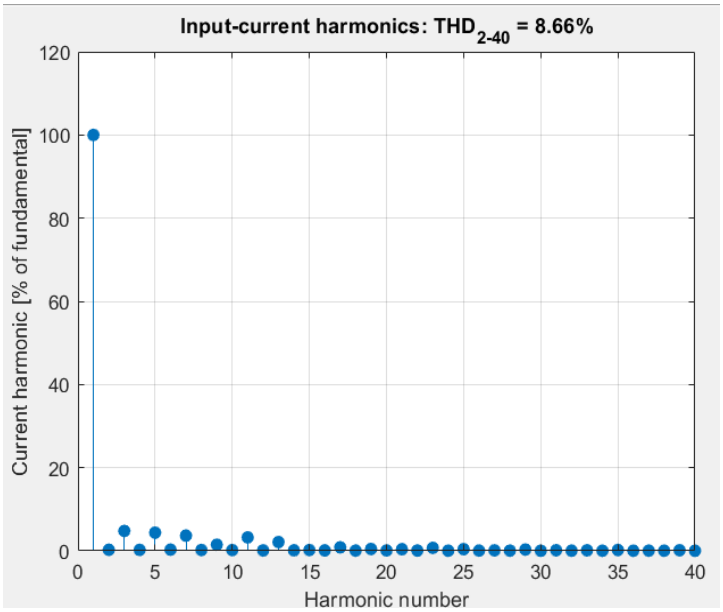


Figure 25 – Input-current harmonic spectrum, 50 %→100 %; THD(2–40) = 8.66 %.

Figure 25 shows how the spectrum changes under pulsing. The dominant seventh/eleventh pattern of the static cases is replaced by a broader, lower comb: third at about 5 %, fifth at about 4.5 %, seventh at about 4 %, eleventh at about 3.3 %, with small components extending to the thirteenth. The pulsed load has injected low-order harmonics, but all of them remain below 5 % of the fundamental.

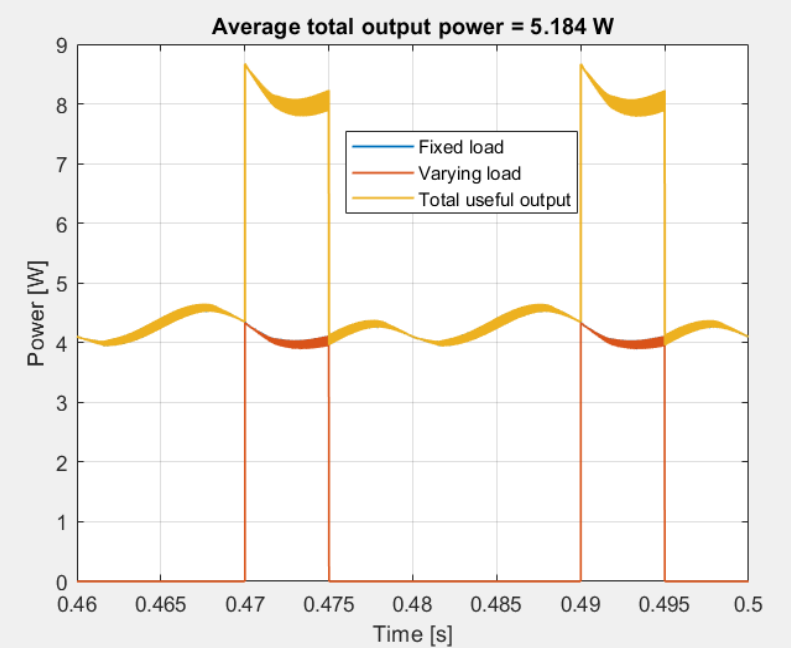


Figure 26 – Output power: fixed load, pulsed load and total useful output; average 5.184 W.

Figure 26 shows what the DC side is actually doing, and makes the magnitude of the disturbance explicit. The fixed load draws a steady 4.0–4.6 W, modulated by the double-line-frequency ripple of the DC bus. Twice per cycle the pulsed load switches on for 5 ms and the total useful output jumps to about 8.7 W before decaying to 8.0 W during the pulse, then drops back. The average total output power over the window is 5.184 W. The contrast between this violently discontinuous output and the smooth sinusoidal input of Figure 24 is the clearest single illustration of the isolating action of the PFC stage.

```
Analysis window: 0.460000 to 0.500000 s (2.000 line cycles)
-------------------------------------------------------------
Input voltage RMS                    =      35.3553 V
Input current RMS                    =     0.176214 A
Average input power                  =       6.2065 W
Input apparent power                 =       6.2301 VA
True power factor                    =     0.996213
Fundamental displacement PF          =     0.999988
V-I fundamental phase difference     =    -0.285877 deg
Current distortion factor I1/Irms    =     0.996225

Input-current THD, harmonics 2-40    =     8.659675 %
Input-current residual THD           =     8.707752 %
Input-voltage THD, harmonics 2-40    =     0.000000 %
Input-voltage residual THD           =     0.000184 %

Average output voltage               =      12.0225 V
Minimum / maximum output voltage     =      11.6094 / 12.6897 V
Fixed-load power                     =      4.17968 W
Varying-load power                   =      1.00466 W
Total useful output power            =      5.18434 W
Efficiency                           =    83.530706 %
Varying-load calculation             = V(N004,COM) * I(R_VARYING)
Output-voltage endpoint change       =     -0.00574 V (-0.0477439 %)
```

Figure 27 – Numerical summary of the analysis window, 50 %→100 %.

Figure 27 breaks the result down: 0.1762 A RMS input current, 6.21 W input power, 6.23 VA apparent power, displacement factor 0.999988, distortion factor 0.9962. The output is composed of 4.18 W of fixed-load power and 1.00 W of pulsed-load power, giving 5.18 W of total useful power and an efficiency of 83.53 %. The output voltage averages 12.02 V and swings between 11.61 V and 12.69 V – a peak-to-peak excursion of about 1.1 V, or ±4.5 %, which is the price paid on the DC side for keeping the AC side clean. The endpoint drift is −0.048 %, so the loop returns to the same operating point every cycle.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.17555	100
2	3e-09	8.4853e-09	0.00046866	0.26697
3	4.4813e-09	1.2675e-08	0.0084036	4.7871
4	8.3665e-09	2.3664e-08	0.00040251	0.22929
5	3.2338e-09	9.1465e-09	0.0076747	4.3718
6	6.4778e-09	1.8322e-08	0.00051797	0.29506
7	3.403e-09	9.6253e-09	0.0064345	3.6654
8	4.3478e-09	1.2298e-08	0.00035579	0.20268
9	7.999e-09	2.2625e-08	0.0025974	1.4796
10	5.8711e-09	1.6606e-08	0.00032614	0.18578
11	1.0663e-09	3.016e-09	0.0056874	3.2398
12	2.8969e-09	8.1937e-09	0.00019553	0.11138
13	4.2961e-09	1.2151e-08	0.0037172	2.1175
14	6.3888e-09	1.807e-08	0.00023072	0.13143
15	5.2802e-09	1.4935e-08	0.00034785	0.19815
16	9.8023e-09	2.7725e-08	0.00019728	0.11238
17	2.3845e-09	6.7443e-09	0.0014729	0.83904
18	7e-09	1.9799e-08	0.00012676	0.072208
19	3.5517e-09	1.0046e-08	0.00077164	0.43956
20	3.6379e-09	1.0289e-08	0.00017054	0.097149

Figure 28 – First 20 harmonics, 50 %→100 %.

Figure 28 gives the numerical spectrum: a fundamental of 0.17555 A with the largest harmonic currents of 8.4 mA at the third, 7.7 mA at the fifth and 6.4 mA at the seventh. The appearance of measurable even harmonics

(0.27 % at the second) is the numerical fingerprint of the pulsed load breaking the half-wave symmetry of the current.

10.6.2 Step from 100 % to 150 %

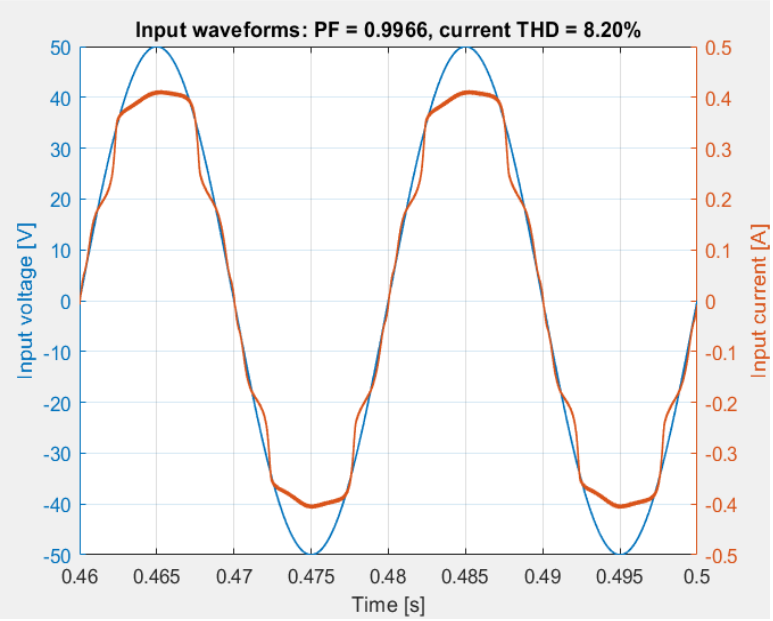


Figure 29 – Input waveforms, 100 %⟶150 % load step; PF = 0.9966, current THD = 8.20 %.

Figure 29 shows the same test at a higher base load. The current peak has risen to about 0.41 A and the crest of the waveform is noticeably flatter than in Figure 24 – the converter is closer to its power limit and the control loop clips the top of the sine. Despite the 50 % load step the power factor is 0.9966, marginally better than the previous case because the fundamental is larger relative to the same disturbance.

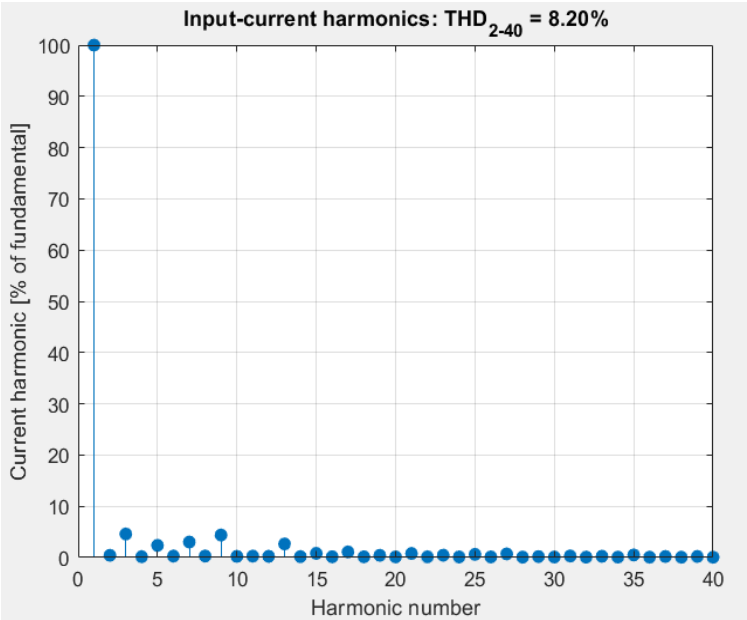


Figure 30 – Input-current harmonic spectrum, 100 %⟶150 %; THD(2–40) = 8.20 %.

Figure 30 shows a spectrum of the same family as Figure 25 – third at about 4.6 %, ninth at about 4.4 %, thirteenth at about 2.7 % – with a slightly lower total distortion of 8.20 %. The shift of weight towards the ninth and thirteenth harmonics reflects the flattened crest visible in the waveform.

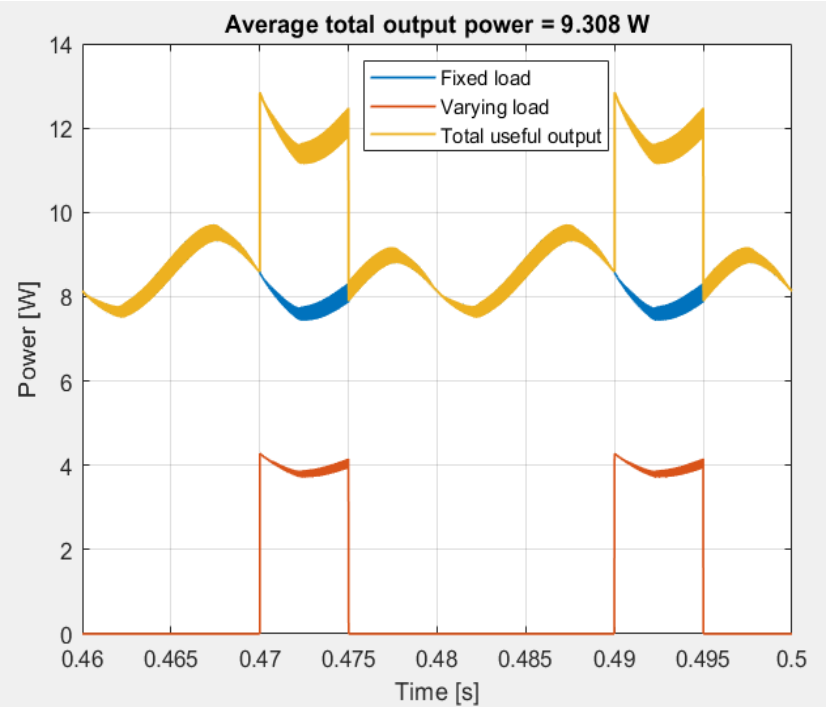


Figure 31 – Output power: fixed load, pulsed load and total useful output; average 10.308 W.

Figure 31 shows the fixed load at 7.6–9.7 W with the pulsed load adding about 4.2 W for 5 ms twice per cycle, taking the total to a peak of 12.8 W. The average total useful output is 10.308 W, nearly double the previous case.

```
Analysis window: 0.460000 to 0.500000 s (2.000 line cycles)
-------------------------------------------------------------
Input voltage RMS                  =      35.3553 V
Input current RMS                  =     0.309567 A
Average input power                =      10.9075 W
Input apparent power               =      10.9448 VA
True power factor                  =     0.996592
Fundamental displacement PF        =     0.999981
V-I fundamental phase difference   =     0.356286 deg
Current distortion factor I1/Irms  =     0.996611

Input-current THD, harmonics 2-40  =     8.203250 %
Input-current residual THD         =     8.243271 %
Input-voltage THD, harmonics 2-40  =     0.000000 %
Input-voltage residual THD         =     0.000238 %

Average output voltage             =      11.9997 V
Minimum / maximum output voltage   =      11.3403 / 12.9652 V
Fixed-load power                   =      8.33367 W
Varying-load power                 =     0.974185 W
Total useful output power          =      9.30786 W
Efficiency                         =    85.334224 %
Varying-load calculation           = V(N004,COM) * I(R_VARYING)
Output-voltage endpoint change     =      0.00011 V (0.000916691 %)
```

Figure 32 – Numerical summary of the analysis window, 100 %→150 %.

Figure 32 reports 0.3096 A RMS, 10.91 W input, 10.94 VA apparent, displacement factor 0.999981, distortion factor 0.9966, and an output composed of 8.33 W fixed plus 0.97 W pulsed, for 10.31 W total and an efficiency

of 85.33 % – the best of the four dynamic cases, since the converter is operating near its nominal design point. The output voltage averages 11.9997 V within a band of 11.34–12.97 V, and the endpoint drift is 0.0009 %.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.30852	100
2	4.4886e-09	1.2696e-08	0.0013369	0.43334
3	9.2125e-09	2.6057e-08	0.014176	4.5948
4	5.9719e-09	1.6891e-08	0.00054325	0.17608
5	2.3657e-09	6.6913e-09	0.0073175	2.3718
6	2.9856e-09	8.4444e-09	0.0008919	0.28909
7	3.527e-09	9.976e-09	0.0093397	3.0273
8	5.2088e-09	1.4733e-08	0.00090758	0.29417
9	7.5055e-09	2.1229e-08	0.013547	4.3911
10	6.0268e-09	1.7046e-08	0.00071751	0.23257
11	4.0163e-09	1.136e-08	0.00087818	0.28464
12	3.1196e-09	8.8236e-09	0.00071287	0.23106
13	5.774e-09	1.6331e-08	0.0081769	2.6504
14	6.8285e-09	1.9314e-08	0.0005456	0.17685
15	7.1004e-09	2.0083e-08	0.0025133	0.81463
16	5.4671e-09	1.5463e-08	0.00048934	0.15861
17	3.9881e-09	1.128e-08	0.0033685	1.0918
18	3.5481e-09	1.0036e-08	0.00046163	0.14963
19	5.3502e-09	1.5133e-08	0.0013564	0.43966
20	3.5993e-09	1.018e-08	0.00038214	0.12386

Figure 33 – First 20 harmonics, 100 %⟶150 %.

Figure 33 lists the corresponding harmonic currents. Compared with Figure 28 every harmonic has grown in absolute terms, but so has the fundamental, which is why the relative THD is essentially unchanged: the distortion scales with the load rather than being added by it.

10.6.3 Step from 100 % to 200 %

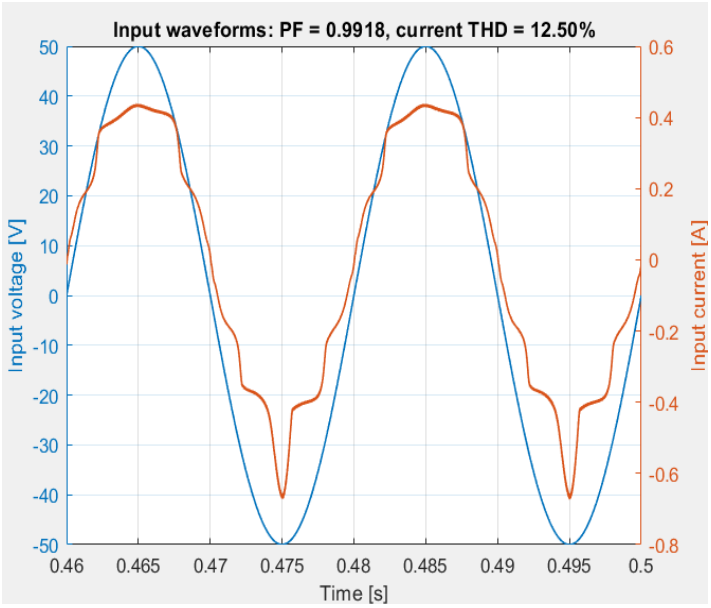


Figure 34 – Input waveforms, 100 %⟶200 % load step; PF = 0.9918, current THD = 12.50 %.

Figure 34 is the first case in which the disturbance becomes clearly visible at the input. The current still follows the voltage over most of the cycle, but a distinct notch and a sharp spike to −0.78 A appear in the negative half-cycle, at the instant the pulsed load switches. The waveform is no longer a clean sine; the power factor has fallen to 0.9918 and the THD has risen to 12.50 %.

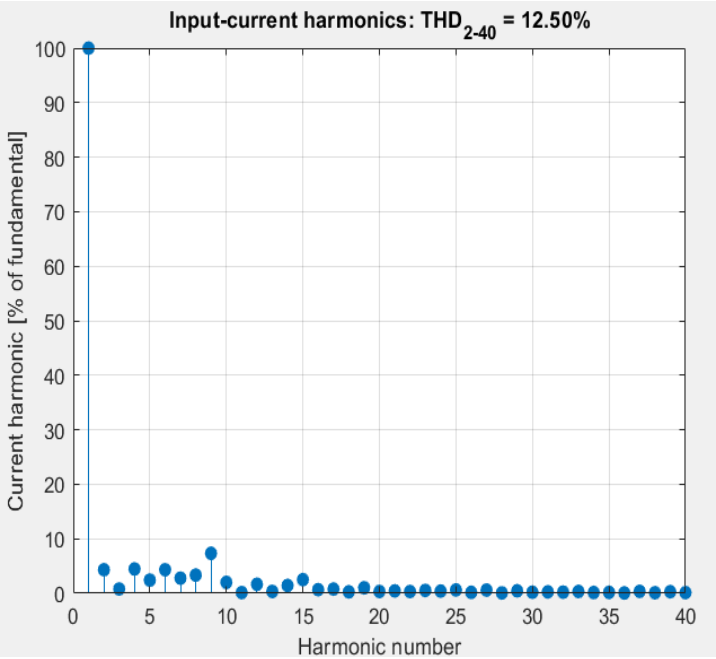


Figure 35 – Input-current harmonic spectrum, 100 %⟶200 %; THD(2–40) = 12.50 %.

Figure 35 shows the consequence in the spectrum: a broad comb of low-order harmonics, with the ninth at about 7 %, the second at about 4.4 % and the fourth, fifth and sixth all between 2 % and 4.5 %. The growth of the even harmonics is significant – it shows that the two half-cycles are no longer identical, because the load pulse is correlated with the line frequency rather than random.

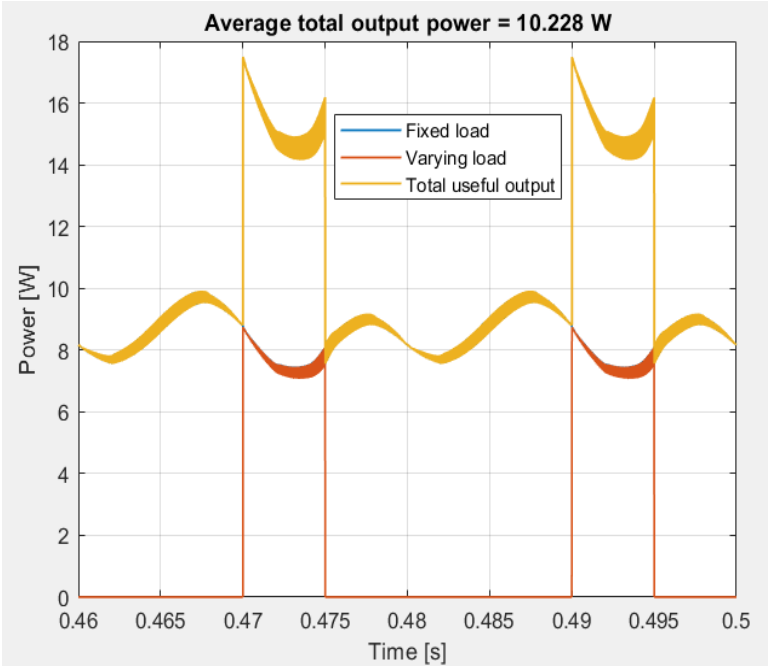


Figure 36 – Output power: fixed load, pulsed load and total useful output; average 10.228 W.

Figure 36 shows why. The pulsed load now adds about 7.5 W on top of a fixed load of 8–9.8 W, so the total output jumps to 17.5 W – roughly double the steady value – and then sags to 14.5 W during the 5 ms pulse as the bus capacitors discharge. The average is 10.228 W. The visible sag of the total-power trace during each pulse is the energy being taken out of the DC bus faster than the slow loop can replace it.

```
Analysis window: 0.460000 to 0.500000 s (2.000 line cycles)
-----------------------------------------------------------
Input voltage RMS                    =      35.3553 V
Input current RMS                    =     0.343096 A
Average input power                  =      12.0307 W
Input apparent power                 =      12.1303 VA
True power factor                    =     0.991794
Fundamental displacement PF          =     0.999973
V-I fundamental phase difference     =     0.423326 deg
Current distortion factor I1/Irms    =     0.991821

Input-current THD, harmonics 2-40    =    12.497685 %
Input-current residual THD           =    12.526794 %
Input-voltage THD, harmonics 2-40    =     0.000000 %
Input-voltage residual THD           =     0.000260 %

Average output voltage               =           12 V
Minimum / maximum output voltage     =      11.0682 / 13.1043 V
Fixed-load power                     =      8.33969 W
Varying-load power                   =      1.88836 W
Total useful output power            =      10.2281 W
Efficiency                           =    85.015990 %
Varying-load calculation             = V(N004,COM) * I(R_VARYING)
Output-voltage endpoint change       =      0.00021 V (0.00174999 %)
```

Figure 37 – Numerical summary of the analysis window, 100 %→200 %.

Figure 37 quantifies the degradation while showing where it does not come from: the displacement factor is still 0.999973 and the phase difference only 0.42°, so the entire loss of power factor is in the distortion factor (0.9918). The input draws 0.3431 A RMS and 12.03 W; the output is 8.34 W fixed plus 1.89 W pulsed, 10.23 W total, at 85.02 % efficiency. The output voltage band has widened to 11.07–13.10 V, about ±8.5 %, while the average stays exactly at 12.0 V.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.34029	100
2	6.2536e-09	1.7688e-08	0.014719	4.3254
3	3.7786e-09	1.0688e-08	0.0026995	0.7933
4	4.4405e-09	1.256e-08	0.015182	4.4616
5	8.5813e-09	2.4272e-08	0.0082401	2.4215
6	3.9618e-09	1.1206e-08	0.014691	4.3172
7	4.1056e-10	1.1613e-09	0.0094628	2.7808
8	1.6979e-09	4.8024e-09	0.01135	3.3355
9	7.6704e-09	2.1695e-08	0.024962	7.3354
10	3.1475e-09	8.9025e-09	0.0068104	2.0014
11	4.5875e-09	1.2975e-08	0.00045281	0.13307
12	9.2278e-09	2.61e-08	0.005623	1.6524
13	4.4629e-09	1.2623e-08	0.0011682	0.34329
14	1.868e-08	5.2836e-08	0.004891	1.4373
15	8.3039e-09	2.3487e-08	0.0085141	2.502
16	3.2852e-09	9.292e-09	0.0022713	0.66746
17	4.4487e-09	1.2583e-08	0.0026523	0.77944
18	9.8578e-09	2.7882e-08	0.00091442	0.26872
19	7.8218e-09	2.2123e-08	0.0034445	1.0122
20	5.4847e-09	1.5513e-08	0.0011476	0.33724

Figure 38 – First 20 harmonics, 100 %→200 %.

Figure 38 shows the even-harmonic content rising to levels comparable with the odd harmonics, which is the tabular counterpart of the asymmetric waveform of Figure 34.

10.6.4 Step from 100 % to 400 % – the limit case

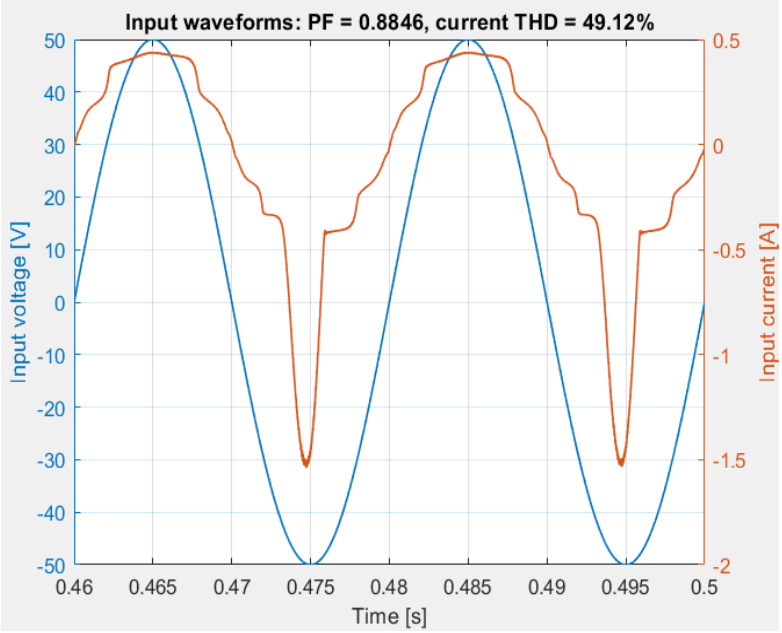


Figure 39 – Input waveforms, 100 %→400 % load step; PF = 0.8846, current THD = 410.12 %.

Figure 39 shows the converter driven beyond its useful operating envelope. A quadrupling of the load twice per cycle produces a deep, wide notch in the negative half-cycle, where the current plunges to −1.5 A, roughly three times the normal peak, and a visibly stepped, non-sinusoidal shape elsewhere. The power factor falls to 0.8846 – below the 0.95 target – and the THD reaches 410.12 %.

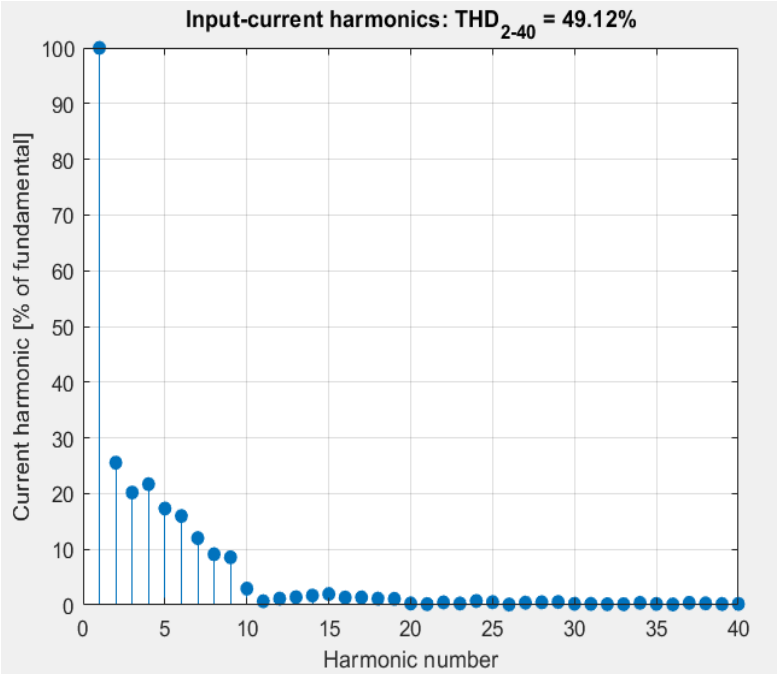


Figure 40 – Input-current harmonic spectrum, 100 %→400 %; THD(2–40) = 410.12 %.

Figure 40 shows a spectrum quite unlike all the previous ones: a monotonically decaying series in which the second harmonic reaches 25 %, the third 20 %, the fourth 22 %, the fifth 17 % and the sixth 13 %, with significant content out to the tenth. Even and odd harmonics are now of comparable size – the current waveform has effectively lost its half-wave symmetry, because the 5 ms load pulse is a large disturbance occurring at twice the line frequency.

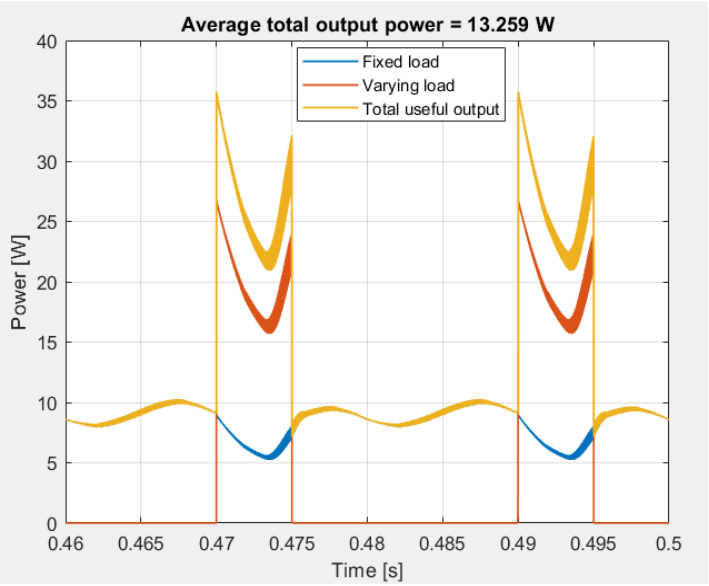


Figure 41 – Output power: fixed load, pulsed load and total useful output; average 13.259 W.

Figure 41 explains the severity of the case. During each pulse the total useful output leaps from about 9 W to 35.7 W – close to four times the nominal rating – and then collapses to about 21 W within the 5 ms pulse, while the fixed-load power visibly sags from 9 W to 5.5 W because the bus voltage itself is falling. The average output is 13.259 W. The bus capacitance simply cannot buffer a disturbance of this size, so the deficit is passed through to the input current.

```
Analysis window: 0.460000 to 0.500000 s (2.000 line cycles)
-------------------------------------------------------------
Input voltage RMS                  =      35.3553 V
Input current RMS                  =     0.506488 A
Average input power                =      15.8401 W
Input apparent power               =      17.9071 VA
True power factor                  =     0.884575
Fundamental displacement PF        =     0.999743
V-I fundamental phase difference   =    -1.299347 deg
Current distortion factor I1/Irms  =     0.884802

Input-current THD, harmonics 2-40  =    49.123617 %
Input-current residual THD         =    49.135595 %
Input-voltage THD, harmonics 2-40  =     0.000000 %
Input-voltage residual THD         =     0.000263 %

Average output voltage             =      11.9976 V
Minimum / maximum output voltage   =      9.52086 / 13.3399 V
Fixed-load power                   =      8.37496 W
Varying-load power                 =      4.88398 W
Total useful output power          =      13.2589 W
Efficiency                         =    83.704754 %
Varying-load calculation           = V(N004,COM) * I(R_VARYING)
Output-voltage endpoint change     =     -0.01143 V (-0.0952693 %)
```

Figure 42 – Numerical summary of the analysis window, 100 %→400 %.

Figure 42 gives the final numbers, and they contain the most important diagnostic of the whole study: the fundamental displacement factor is still 0.999743, with a phase difference of only −1.30°. Even in this extreme case the PFC stage keeps the fundamental current perfectly aligned with the voltage; the entire loss of power factor (distortion factor 0.8848) is harmonic distortion, not reactive power. The input draws 0.5065 A RMS and 15.84 W for 17.91 VA; the output is 8.37 W fixed plus 4.88 W pulsed, 13.26 W total, at 83.70 % efficiency. The output voltage band is 10.52–13.34 V – a dip of more than 2.5 V below nominal – with an endpoint drift of −0.095 %.

First 20 harmonic rows:

Harmonic	V_RMS_V	V_pct_of_fundamental	I_RMS_A	I_pct_of_fundamental
1	35.355	100	0.44814	100
2	9.7721e-10	2.764e-09	0.11445	25.54
3	7.1817e-09	2.0313e-08	0.090471	20.188
4	1.4294e-09	4.0429e-09	0.097207	21.691
5	8.0304e-09	2.2713e-08	0.07757	17.309
6	4.2706e-09	1.2079e-08	0.071532	15.962
7	4.7135e-09	1.3332e-08	0.053819	12.009
8	8.7617e-09	2.4782e-08	0.040834	9.1118
9	1.0317e-08	2.918e-08	0.038368	8.5615
10	4.1126e-09	1.1632e-08	0.013165	2.9376
11	1.8504e-09	5.2337e-09	0.0029898	0.66716
12	1.1292e-08	3.1938e-08	0.0051349	1.1458
13	3.4931e-09	9.88e-09	0.0062854	1.4026
14	1.0191e-08	2.8824e-08	0.0075975	1.6953
15	8.8237e-09	2.4957e-08	0.0088641	1.978
16	6.8846e-09	1.9473e-08	0.0059581	1.3295
17	6.3317e-09	1.7909e-08	0.0061121	1.3639
18	5.9272e-09	1.6765e-08	0.0050858	1.1349
19	6.0698e-09	1.7168e-08	0.0049498	1.1045
20	7.3822e-09	2.088e-08	0.0012909	0.28805

Figure 43 – First 20 harmonics, 100 %→400 %.

Figure 43 closes the simulation set with the numerical spectrum of the limit case, in which the low-order harmonic currents are of the order of a hundred milliamperes against a fundamental of roughly 450 mA. Comparing this table with Figure 15 shows the whole span of the study: the same converter, the same controller, and a harmonic content that grows by two orders of magnitude when – and only when – the load transient becomes comparable to the energy stored in the DC bus.

11. Comparative Analysis

Table 1 collects the static cases, in which the output power was held approximately constant and only the character of the load was changed.

Case	PF	Displacement factor	THD 2–40 [%]	I_rms [A]	P_in [W]	P_out [W]	η [%]
No PFC (reference)	0.5698	0.9800	139.77	0.5623	11.33	9.08	80.13
PFC – resistive	0.9976	0.997562	6.95	0.2775	9.79	8.33	85.12
PFC – inductive	0.9975	0.997521	7.01	0.2745	10.68	8.24	85.07
PFC – capacitive	0.9975	0.997547	6.97	0.2751	9.70	8.30	85.59

Table 1 – Effect of load character at constant power (analysis window 0.46–0.50 s).

The three corrected cases differ by 0.01 % in power factor, 0.06 percentage points in THD and 0.5 percentage points in efficiency. Against the 75 % improvement in power factor and the twenty-fold reduction in THD obtained by enabling the PFC stage, these differences are negligible. The research hypothesis – that the PFC stage decouples the character of the load from the mains – is therefore confirmed for static loads.

Table 2 collects the dynamic cases, in which the load character was held fixed and only the magnitude of the repeated load step was changed.

Load step	PF	THD 2–40 [%]	P_out avg [W]	V_out min/max [V]	η [%]
50 % → 100 %	0.9962	8.66	5.18	11.61 / 12.69	83.53
100 % → 150 %	0.9966	8.20	10.31	11.34 / 12.97	85.33
100 % → 200 %	0.9918	12.50	10.23	11.07 / 13.10	85.02
100 % → 400 %	0.8846	410.12	13.26	10.52 / 13.34	83.70

Table 2 – Effect of load-step magnitude at fixed load character.

Three observations follow from Table 2. First, for load steps up to and including a doubling of the output power, the power factor stays above 0.99 and the THD below 12.5 %, comfortably inside the 0.95 target. Second, the degradation is strongly non-linear: going from a 2× step to a 4× step multiplies the THD by four. Third, and most important, the displacement factor remains between 0.99974 and 0.99999 in every single case – the control loop never loses phase alignment with the mains. What fails at 400 % is not the correction principle but the energy buffer: the bulk capacitance is sized for the nominal load, and once the pulse energy approaches the stored energy, the bus voltage collapses (down to 10.52 V) and the input current must supply the deficit within the pulse, which is by definition a non-sinusoidal demand.

The two tables together make the distinction that the project set out to establish. The power factor is indifferent to what kind of load is connected; it is sensitive only to how much energy the load demands and how fast it demands it, relative to the storage designed into the converter.

12. Conclusions

- The control law was verified directly at cycle level. Both of our models run at a fixed 100 kHz and in discontinuous conduction; in the PFC model the on-time is held constant to within 0.08 % across the line cycle (3.4296–3.4322 µs), which is precisely the condition that makes the converter emulate a resistance and therefore shapes the input current without any fast current loop.
- Our PFC control law delivers its headline benefit. Enabling the correction raises the power factor from 0.5698 to 0.9976 and reduces the input-current THD from 139.77 % to 6.95 %, while raising the efficiency from 80.13 % to 85.12 % – the last because the RMS input current falls by half for the same delivered power.
- The power factor is independent of load character. Resistive, inductive and capacitive loads at the same power level give 0.9976, 0.9975 and 0.9975, with THD of 6.95 %, 7.01 % and 6.97 %. The research hypothesis is confirmed.
- The power factor is not independent of load dynamics. Repeated load steps of up to 2× keep PF above 0.99, but a 4× step degrades it to 0.8846 with 49.12 % THD. This is a limitation of the energy stored in the DC bus, not of the correction principle.
- The failure mode is distortion, never displacement. In every case measured, including the worst, the fundamental displacement factor is at least 0.9997 and the V–I phase difference never exceeds 1.3°. All observed power-factor loss is harmonic.

- Output regulation holds throughout. The average output voltage remains within 0.2 % of 12 V in every simulation, and the endpoint drift across two line cycles never exceeds 0.2 %; the instantaneous excursion grows with the load step, from ±4.5 % at a 2× step to a 2.5 V dip at a 4× step.

12.1 Recommendations for the laboratory phase

- Build the hardware prototype of the design of Chapter 8 and reproduce the four static cases first, using the agreement (or otherwise) with Table 1 to validate the simulation models before proceeding.
- Sweep the load-step magnitude finely between 2× and 4× to locate the point at which the power factor crosses the 0.95 target; the simulation places it in this interval.
- Measure the recovery time of the output voltage after a single step, which the two-cycle analysis window used here cannot resolve.
- Repeat the 4× case with additional bulk capacitance on the DC bus, to confirm experimentally that the degradation is an energy-storage limitation rather than a control-loop limitation.
- Compare the measured prototype against a BCM variant with a zero-current-detect winding, to quantify experimentally the turn-on loss and RMS-current penalty that the fixed-frequency DCM choice accepts.

Appendix A – Laboratory Session: Power Factor Invariance in a PFC Flyback Converter

This appendix reproduces the guided laboratory session that accompanies the converter as part of the delivered product. It is written for students and is intended to be run directly on the LTspice models of Chapter 8.

Duration: ~15–20 min | Format: guided LTspice simulation | Prerequisites: basic AC power (P, S, PF), DCM flyback operation

A.1 Learning objectives

- State the definition of power factor and distinguish it from just “voltage regulation.”
- Identify how this converter's control scheme (constant switch on-time, slow outer voltage loop) shapes the input current.
- Predict – before running the sim – that PF should stay near-unity regardless of the DC-side load type or a load step, and explain why.
- Run LTspice variants that confirm this experimentally and log the results.

A.2 Session timeline

Time	Segment	What happens
0:00–2:00	Hook	Why power factor matters (harmonics, IEC 61000-3-2, utility penalties).
2:00–5:00	Circuit walkthrough	Trace the PFC block; land on the key idea: constant on-time → current envelope follows Vin(t).
5:00–7:00	Derivation (board)	Ipk(t) = Vin(t)·Ton / Lp → naturally sinusoidal peak-current envelope → PF ≈ 1, independent of load.
7:00–15:00	Guided sim runs	Students run 3–4 LTspice variants (R, R+C, R+L, step-load) and log PF, Vout ripple, Ton behaviour.
15:00–18:00	Discussion	Why doesn't PF care what's on the output? Ties back to fast current loop vs. slow voltage loop.

A.3 Part 1 – Background (2–5 min)

Power factor: PF = P / S = P / (Vrms·Irms). A load can draw the correct average power at the correct RMS current and still have a bad PF if the current is not in phase with, and shaped like, the voltage. A plain diode-bridge-plus-cap front end draws narrow current pulses near the voltage peaks – PF often 0.5–0.6 – which is why PFC exists and why standards like IEC 61000-3-2 mandate it above a few watts.

Point out that PF has two components: displacement (phase shift of the fundamental) and distortion (harmonics from non-sinusoidal shape). This circuit's job is to fix both by making the input current track the input voltage waveform.

A.4 Part 2 – Circuit walkthrough and the key idea (3–5 min)

Walk the signal path: EMI filter (Lf, Cx) → diode bridge → flyback (Lp/Ls, Q1) → output rectifier/cap → opto-isolated feedback (TL431) → slow low-pass (Rfb/Cfb) → PI controller that sets TON_CMD.

Have students find the line in the netlist that sets the switch on-time:

```
BTON_CMD  TON_CMD = limit( Ton0 + Kpton·(Vref − Vfb_slow) + Kiton·∫(Vref − Vfb_slow)dt
, TonMin, TonMax )
```

The crucial detail: Vfb_slow is heavily low-pass filtered (Rfb = 100 kΩ, Cfb = 150 nF → τ = 15 ms), so Ton barely moves within a single line half-cycle (10 ms at 50 Hz). Over one switching period Ton is effectively constant – it only drifts slowly, cycle-to-cycle, to hold Vout at its set point.

Now derive the consequence on the board. With Ton essentially fixed within a switching cycle, the primary peak current each cycle is Ipk(t) ≈ Vin(t) · Ton / Lp. Since Ton and Lp are constant over the cycle, Ipk(t) is directly proportional to the instantaneous rectified line voltage. The peak-current envelope is automatically a scaled copy of |Vin(t)| – i.e. naturally sinusoidal – with no multiplier IC and no line-voltage sensing needed. This is the classic constant-on-time (COT) borderline/DCM PFC technique.

This is the single fact that explains everything the students are about to observe: the current shaping happens cycle-by-cycle at the primary side, driven only by Vin(t), Lp and Ton. The DC-side load (R_LOAD) never enters that equation — it only feeds back, slowly, to set the average value of Ton. Change the load and Ton re-settles to a new average; the shape of the current within each line cycle is untouched.

A.5 Part 3 – Guided simulation runs (7–9 min)

Students duplicate the baseline .asc/.net four times (or use .step) and make one change each run. Everything else – including the .meas power_factor block – stays identical so that the results are directly comparable.

Run	Load config	Netlist change	What to record
1	Resistive (baseline)	Keep R_LOAD as-is: R_LOAD out COM {RloadVal}, RloadVal=17.3	v_rms, i_rms, power_factor, vout_avg, ton_avg
2	Resistive + bulk C (“capacitive” DC load)	Add: Cbig out COM 2200µ (parallel with R_LOAD)	power_factor, vout ripple (vout_max−vout_min), ton_avg vs. Run 1
3	Resistive + series L (“inductive” DC load)	Break R_LOAD into two nodes: Lload out Lmid 2m Rser=0.2 / R_LOAD Lmid COM {RloadVal}	power_factor, transient sag on step, ton_avg vs. Run 1
4	Time-varying load (step)	Replace R_LOAD with: R_LOAD out COM R=if(time<0.3,17.3,8.6) (needs behavioural resistor – or use two R_LOAD in parallel gated by a switch)	power_factor computed separately before/after 0.3 s (edit the FROM/TO window in .meas), vout_avg before/after

Tip for Run 4 (behavioural R_LOAD): LTspice supports a behavioural resistor with R=expression. If that syntax misbehaves in your LTspice version, use two fixed R_LOAD branches in parallel, each gated on/off with a voltage-controlled switch driven by a PULSE source, and sum their currents.

A.6 Results log (fill in during lab)

Run	v_rms	i_rms	power_factor	vout_avg	Notes
1 – Resistive					
2 – R + C					
3 – R + L					
4 – Step load					

A.7 Part 4 – Discussion (2–4 min)

- Compare PF across all four runs. It should sit within ~0.001–0.005 of the baseline in every case – ask students to explain why in terms of the Part 2 derivation, not just "the feedback loop handles it."
- What does change across runs? Vout ripple, ton_avg and the transient shape after the step – all output-side quantities. Ask: why does the input-side PF stay put while these move?
- What would break this? Ask students to predict what happens to PF if Cfb is made much smaller (fast voltage loop) so that Ton starts tracking within a line half-cycle instead of staying constant – have them test it if time allows.
- Real-world caveat: this control scheme needs no line-voltage sense and no multiplier, which is why it is attractive for a low-cost/Arduino-class controller – but it assumes the input voltage magnitude itself is roughly constant run-to-run (it is not ideal for universal 90–265 Vac unless Ton0 is also adjusted with Vin,RMS).